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Machine Learning-Based Analysis of Instagram and Pinterest Content for Fashion Trend Discovery

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Declaration

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May 8, 2026

Abstract

Fashion trend forecasting has traditionally been a top-down process, driven by designers and industry experts with limited input from consumer behaviour. The rise of social media has shifted this dynamic, enabling trends to emerge organically from user-generated content, though this data is unstructured, noisy, and largely unlabelled, making it difficult to analyse systematically. This project investigates whether multimodal social media data, combining Instagram captions and Pinterest images, can be used to detect and predict fashion trends. CLIP (Contrastive Language-Image Pretraining) was used to embed text and visual data into a shared feature space, after which K-means clustering ($k = 8$) identified eight distinct fashion archetypes, including denim, menswear, luxury fashion, and promotional content. Temporal analysis was then performed by aggregating cluster membership monthly, with momentum-based measures applied to detect emerging trends. The identified clusters aligned well with documented 2015–2016 fashion movements reported in industry publications, suggesting the pipeline is effective at capturing real-world trend signals. For forecasting, linear regression was compared against a naïve baseline; the naïve model outperformed regression, reflecting the high volatility and short-lived nature of social media trends, which are heavily shaped by platform dynamics and influencer activity. These findings suggest that while multimodal social media data is well-suited to trend detection and retrospective analysis, reliable forecasting remains a significant challenge. An important distinction for both researchers and practitioners considering data-driven approaches to fashion intelligence.

Keywords: Multimodal Learning, Fashion Trend Detection, Social Media Analysis, CLIP Embeddings, Temporal Clustering

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Program code link: <https://csgitlab.reading.ac.uk/rp011422/cs3ip-fashion-trend-discovery>

Web Application link: <https://fashion-trend-discovery-dashboard.streamlit.app/>

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List of Abbreviations

AI	Artificial Intelligence
ARIMA	AutoRegressive Integrated Moving Average
CLIP	Contrastive Language–Image Pretraining
ECCV	European Conference on Computer Vision
EDA	Exploratory Data Analysis
GDPR	General Data Protection Regulation
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
K-Means	K-Means Clustering Algorithm
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
NLP	Natural Language Processing
OOTD	Outfit Of The Day
PCA	Principal Component Analysis

1 Introduction

1.1 Background

Traditionally, experts decided what was fashionable. Designers, buyers, and forecasters would study runway collections or analyse editorial magazines (Vogue, Elle, Vanity Fair) to predict what people would eventually wear. Despite these methods working well for established luxury brands, the problem is that human curation simply does not scale. As consumer preferences shift rapidly, and traditional market research often falls behind. Nevertheless, social media has single handedly reshaped this demographic, platforms like Instagram and Pinterest now operate as massive repositories of user-generated fashion data. This invites data analytics into the equation, where analysts can finally track visual patterns and engagement metrics on an enormous scale [Wang et al., 2015]. In contrast to the traditional hierarchical trend propagation, where designers and fashion houses dictate the direction, social media enables a decentralised, participatory model for trend formation. Individuals who have built a substantial following on social media are called “Influencers”, they shape the audience’s preferences through their content and now play a dominant role in this ecosystem. Content such as visual posts, captions and hashtags, often promotional in nature, reach large audiences and reflects a broad spectrum of fashion expression within a highly user-driven social media environment.

The analogue to digital shift makes deep data analysis possible, researchers increasingly feed these massive datasets into machine learning algorithms to track style patterns as they develop in real time [Liu and Luo, 2020; Li and Zhou, 2024]. However, social media is notoriously noisy, the data is vast and highly heterogenous, therefore when finding meaning within that noise without relying on rigid, predefined categories requires advanced technical approaches.

Machine learning offers a promising approach to fashion trend analysis, processing large volumes of multimodal data automatically allows systems to identify hidden structures and repeating patterns. Ultimately, this provides a highly systematic, exploratory alternative to traditional creative creation.

1.2 Problem Statement

Fashion forecasting can be a challenging task, mainly because it is very intuitive and dynamic. In the past, trend forecasting was a highly subjective process based on expert opinion and historical analysis of the industry, which restricts scalability and responsiveness to real time consumer demand. Can analysis continue to use top-down methods when trend formation has become so heavily decentralised? The evidence surrounded this question disagrees, the established methods have now become decreasingly able to reflect the user-led nature of the style cycle.

Instagram and Pinterest are two of the main social media platforms that have transformed how these trends were generated. The involvement of these large-scale collective user engagements now drives the adoption of fashion, rather than a small number of luxury fashion houses. This reveals a large amount of multimodal data, creating a constant flow of images, descriptions, and hashtags that capture different stylistic preferences. However, this data is difficult to analyse as they originate from a highly variable and unstructured environment. The way algorithms drive engagement can also lead to bias which easily undermines the validity of patterns. [Sharma et al., 2022].

Supervised learning involves data that has been labelled with predefined input/output relationships Digitalyalo (2021). Currently, most machine learning models have tackled this issue using supervised learning, as they require fixed categories and large amounts of training data. Such categorisation of emergent styles inevitably limits what an algorithm can see. How can a system detect a developing style if it is not among the set of training labels? It often can't. Since these systems are confined to fixed labels, they limit the flexibility and the usefulness of these systems in evolving digital environments [Kalyanam and Quezada, 2021]. When trying to predict future trends, it becomes even more complicated, because fashion is shaped by non-linear external influences and algorithmic biases that can change the actual picture of trend diffusion.

These challenges require a new solution, new technique that can handle large volumes of social media data without imposing a strict classification. This paper presents a multimodal machine learning framework that applies unsupervised learning

techniques to uncover latent fashion trends across Instagram and Pinterest. By combining text and images, this approach enables data driven pattern discovery without predefined labels, while also examining the feasibility and limitations of trend forecasting in a highly dynamic digital ecosystem.

1.2.1 Research Questions

This research examines whether machine learning can truly help us understand fashion trends by analysing the mix of images and text found on social media. By looking at both the potential and the clean boundaries of these technologies, the hope is to define the current state of the field. The study addresses several specific questions about how we track style in a digital environment:

- 1) To what extent can unsupervised machine learning, using multimodal embeddings (e.g., CLIP), identify meaningful and interpretable fashion trends from social media data (images and text) ?
- 2) How well do these identified trends match the real-world fashion movements, as observed in industry sources (e.g., Vogue reports from 2015-2016)
- 3) How effectively can simple forecasting approaches (e.g., Linear regression) capture short-term trend dynamics, and how do they perform relative to the naïve baseline models?
- 4) What limitations arise when applying machine learning to social media data for fashion trend analysis, particularly in relation to data noise, platform bias (e.g., Recommendation systems), and commercial or promotional content?
- 5) To what extent can this approach offer a faster, more data-heavy alternative to the traditional ways experts have always forecasted fashion?

1.3 Aims and Objectives

The primary project goal is simple, the creation of a machine learning model to discover and analyse fashion trends from multimodal social media content. This work focuses on unsupervised discovery and the accuracy of short-term predictions. Social media is notoriously noisy, can an algorithm find real signals in all that chaos?

To deliver on this aim, the project has the following objectives:

- 1) To construct a dataset combining Instagram captions with Pinterest images to capture diverse user-generated content.

- 2) Process these text and visual information using Contrastive Language-Image Pre-training (CLIP). This pretrained model generates multimodal embeddings so we can compare different media directly.
- 3) To categorise without predefined rules. By clustering the embedded data, we identify meaningful fashion archetypes entirely from scratch. A completely label-free method, which is then used to track how cluster distributions change over time to see which styles are growing or fading.
- 4) To analyse temporal dynamics by collecting cluster distributions over time, which enables the identification of emerging and declining trends.
- 5) To implement and evaluate baseline forecasting models, assessing the ability to capture short term trend dynamics using appropriate error metrics (e.g., MAE, MAPE) and comparisons against naïve baselines.
- 6) To validate the extracted trends by comparing model outputs with real world fashion trends reported in industry reports (e.g., Vogue).
- 7) To critically evaluate the limitations of applying machine learning to social media data, with specific emphasis on data noise, platform-induced bias (e.g., Recommendation systems), and the influence of commercial or promotional content.

1.4 Solution Approach

This project approached the analysis of fashion trends through exploratory, data driven techniques applied directly to multimodal social media content. The architecture is modular, it combines discrete steps for data collection, representation, clustering, temporal analysis, and forecasting into a single pipeline. How do we effectively map distinct data types into a unified feature space? A multimodal representation model forms the core of this system to solve this. A pretrained vision language model, specifically called CLIP, is applied to process raw images and their accompanying captions from platforms like Instagram and Pinterest [Radford et al., 2021]. Models of this type appear particularly well-suited for massive, unstructured data pools with weak labels. By relying on noisy text annotations, they still manage to learn highly meaningful representation from the underlying media [Saharia et al., 2021].

After feature extraction is complete, the system applied unsupervised clustering to group the data into potential categories. These form distinct fashion archetypes,

because they are derived entirely from the raw data without any prior labelling, they allow a truly open-ended exploration of emerging styles. To understand trend dynamics, we aggregate the cluster distributions over specific time intervals. Tracking which styles gain or lose traction makes it possible to quantify exactly how fashion evolves across various social platforms. The resulting time series is then analysed using straightforward modelling techniques to generate short-term forecasts. Model performance is evaluated using standard error metrics such as Mean Absolute percentage Error (MAPE) and compared against naïve baseline models. This establishes a benchmark for minimum expected accuracy whilst highlighting the limitations of simple forecasting approaches.

An interactive dashboard visualises the cluster properties alongside the time series and forecasts. This interface supports both quantitative assessment and deeper qualitative review. It also exposes underlying complications within the data itself. Noise, algorithmic platform bias, and heavy advertising presence all become immediately visible during manual analysis.

2 Literature Review

2.1 State-of-the-Art in Fashion Trend Analysis Using Social Media

The rapid expansion of visual social media platforms has significantly altered how fashion trends emerge, circulate, and evolve. Platforms such as Instagram and Pinterest function as large-scale, continuously updated repositories of user-generated fashion content, where images, captions, hashtags, and engagement patterns collectively reflect stylistic preferences and forms of consumer attention. As a result, these platforms have become increasingly important data sources for machine learning-based fashion analysis. Unlike traditional top-down models of trend dissemination, where designers and industry experts dictate direction, social media enables a decentralised and participatory process in which trends emerge organically from collective user behaviour.

Traditionally, fashion trend analysis relied on expert judgement, manual curation, and retrospective market research. While these approaches remain influential within the fashion industry, they are often constrained by limited scalability and weak

responsiveness to rapidly changing consumer behaviour. In contrast, machine learning enables the systematic analysis of large volumes of social media data, making it possible to identify recurring visual and textual patterns across diverse forms of fashion content. Prior research demonstrates that analysing social media imagery alongside associated metadata can reveal meaningful aesthetic themes, garment preferences, and stylistic shifts, in some cases before they are formally recognised by industry commentators or commercial actors (Liu and Luo, 2020; Said and Alharbi, 2025). However, much of this work relies on supervised learning approaches, where models are trained using labelled datasets or predefined fashion categories. While effective in controlled environments, these methods embed assumptions about what constitutes a “trend”, potentially reinforcing existing biases and limiting sensitivity to novel or weakly emerging styles.

In response to these limitations, recent research has increasingly explored unsupervised and representation learning approaches, which aim to uncover latent structure directly from data without reliance on predefined labels. This paradigm is particularly well suited to fashion trend analysis, where the objective extends beyond classification to include the discovery of evolving aesthetic groupings. Studies in unsupervised visual representation learning demonstrate that meaningful groupings can be learned without explicit annotation, supporting more exploratory forms of analysis (Caron et al., 2018). More recently, advances in multimodal learning have further expanded this capability. Vision-language models enable images and text to be embedded within a shared semantic space, allowing visual and textual signals to be analysed jointly. Models such as CLIP illustrate the effectiveness of this approach by learning transferable representations from large-scale, weakly aligned image–text data (Radford et al., 2021; Saharia et al., 2021).

Beyond technical developments, the use of machine learning on social media data raises important ethical and societal concerns, including issues of bias, privacy, and unequal representation. These challenges highlight the need to consider methodological effectiveness alongside fairness and responsible data use (Binns and Veale, 2023). Overall, the literature reflects a transition from structured, label-dependent approaches towards more flexible, data-driven methods, motivating the

adoption of exploratory techniques for fashion trend discovery in complex and dynamic social media environments.

2.2 Machine Learning for Fashion Trend Prediction

Machine learning has become a central methodological tool for analysing and forecasting fashion trends, particularly within social media environments characterised by large-scale, high-dimensional, and multimodal data. Fashion-related content on platforms such as Instagram and Pinterest typically combines visual elements with textual information, making it well suited to models capable of integrating heterogeneous data sources. Early research demonstrates that visual features extracted from social media images can act as strong indicators of fashion trends, with large-scale image mining enabling the identification of recurring patterns related to garments, colours, and stylistic attributes (Liu and Luo, 2020). Complementary studies highlight the importance of incorporating textual metadata, showing that combining captions and hashtags with image representations improves trend prediction performance (Said and Alharbi, 2025). Together, these findings emphasise the value of multimodal analysis in capturing the complexity of fashion trends.

Despite these advances, most existing systems rely on supervised learning frameworks, where models are trained on labelled datasets that define fashion categories in advance. Such approaches are widely used due to their interpretability and strong empirical performance, particularly when large, annotated datasets are available. Systems such as Social4Fashion exemplify this paradigm, classifying fashion images into predefined categories before aggregating results to infer trend dynamics (Li and Zhou, 2024). However, this reliance on labelled data introduces several limitations. Label creation is often labour-intensive and subjective, and predefined categories impose rigid structures on data that is inherently fluid and culturally contextual. As a result, supervised models may struggle to detect emerging or unconventional trends that do not conform to established classifications.

Broader research on social media trend prediction reinforces these concerns. Studies show that topic popularity is highly dynamic and influenced by latent factors that are difficult to capture using predefined labels (Kalyanam and Quezada, 2021). Consequently, while supervised models may perform well in structured or short-term

prediction tasks, they are less effective at identifying early-stage or weak signals of emerging trends. In response, there has been increasing interest in unsupervised and representation learning approaches, which enable patterns to be discovered directly from data without explicit labelling. When combined with multimodal techniques, these approaches provide more flexible and contextually rich representations of fashion content. Overall, the literature suggests a shift towards exploratory methodologies that prioritise pattern discovery over rigid classification, particularly in dynamic social media environments.

2.3 Positioning of the Proposed Project Within Existing Literature

The proposed project is positioned between multimodal representation learning, unsupervised pattern discovery, and data driven trend analysis within social media. As mentioned in sections 2.1 and 2.2, existing research applied within fashion trend analytics has been largely divided between supervised learning methods which prioritises structured classification, and unsupervised learning methods which emphasises a more flexible and exploratory approach to pattern discovery. Social4Fashion (Li and Zhou, 2024) demonstrate a supervised learning approach which revealed strong results within controlled settings and relies on the theory that fashion trends are always stable and are known in advance. This theory is incompatible with the noisy and highly individualised nature of fashion expression on social media. In addition, unsupervised approaches offer flexibility but introduces challenges related to interpretability and evaluation. The proposed work builds upon this shift in existing literature by adopting a multimodal, unsupervised framework which captures latent structures within fashion data whilst remaining sensitive to the dynamic nature of social media trends.

Specifically, this project employs vision language models to embed visual and textual data within the same space, enabling the multimodal analysis of images and associated metadata. This agrees with recent advancements in multimodal training, where models such as CLIP have shown the ability to learn transferable representations from large scale, weakly aligned image/text data (Radford et al., 2021). However, instead of applying such models within a heavily descriptive or classification based setting, this projects approach extends the use towards

exploratory trend discovery and temporal analysis. By combining embedding techniques with unsupervised clustering, this project aims to identify coherent fashion groupings without reliance on predefined categories, addressing the key limitation with supervised methods. Furthermore, this project integrates temporal aggregation and forecasting which introduces an additional layer of analysis that is not consistently addressed within existing literature, whereas recent studies focus on static pattern identification rather than the evolution of trends over time.

Within this context the proposed project contributes by creating an intersection between representation learning and practical trend forecasting. While previous research confirms the effectiveness of multimodal embeddings in capturing relationships within fashion data, few studies evaluate how they can support predictive tasks in dynamic environments. The supplementation of baseline comparisons and forecasting evaluation metrics within this project provides a structured framework for assessing model performance, which aids in addressing the challenge of valuating unsupervised systems in the absence of labelled ground truth. However, this approach also reflects the inherent complications identified in the literature, as it introduces limitations within interpretability and validation. Therefore, this project is positioned not as a definitive predictive solution, but as an exploratory framework that evaluates the application of multimodal machine learning in fashion trend analysis whilst simultaneously critically analysing the methodological constraints as discussed.

2.4 Relevance of Existing Work to the Intended Application

The design of this application has directly been based on existing literature, system architecture matters. Instagram and Pinterest offer massive multimodal datasets, visual content is tied directly to textual metadata like captions, hashtags, and user interactions, which provides the scale and contextual depth required for analysing modern fashion dynamics as Liu and Luo (2020) and Said and Alharbi (2025) have pointed out. This supports their use as primary sources as integration is key. Models that blend visual and textual features simply produce much better representation than those trapped within a single modality. Therefore, this project has adopted a unified framework, images and text are in a shared embedded space, allowing the system to capture stylistic trends alongside the actual meaning behind how humans discuss fashion trends online. Existing systems provide a clear guideline, Supervised

approaches like Social4Fashion favour supervised learning however they are limited when applied to dynamic environments. Classified categories impose several restrictions, as the identification of emerging or unconventional trends that do not belong to existing trends become unfeasible. Kalyanam and Quezada (2021) reinforce this by showing that early-stage trends are subtle, diffuse, and shaped by latent factors that evade rigid classification frameworks. This provides a clear rationale for the project for unsupervised and exploratory approaches.

Caron et al. (2018) demonstrated that meaningful patterns emerge straight from the data without any manual annotation. However, this shift introduces some practical challenges as validation gets tricky. This project addresses the ethical considerations regarding bias, data representation, and uneven user participation across social media platforms. Overall, the proposed application takes works from established research and explicitly addresses the historical blind spots. The result is a theoretical framework that handles the messy variability of real world fashion analysis.

2.5 Social, Ethical, and Legal Considerations in Social Media-Based Machine Learning

Feeding social media data into machine learning models for fashion trend analysis introduces immediate ethical and legal complications. Instagram and Pinterest supply enormous amounts of publicly accessible content but scraping all that user-generated content for trend analysis immediately triggers questions about privacy and consent. Algorithmic bias remains a primary concern throughout the literature, it often stems from unrepresented datasets or assumptions created within model design and data labelling. Binns and Veale (2023) demonstrate that machine learning models operating within social media context often reinforce existing inequalities or disproportionately amplify dominant groups. Commercially promoted or culturally dominant styles can easily dominate the results, and less visible forms of expression end up marginalised. Supervised frameworks worsen this situation, as predefined categories tend to reflect strict industry conventions rather than the reality of everyday fashion styles. Thus, this questions how we define and interpret trends, rather than simply reflecting underlying data inconsistencies. Data governance and privacy introduce challenges, much of which online content is public, users however rarely expect their personal posts to be harvested and processed for academic research or predictive modelling.

Legal frameworks like the General Data Protection Regulation (GDPR) enforce principles of data minimization and purpose limitation that apply just as strictly within university contexts. Transparency in sourcing, processing, and storing data is a necessity. Methodological approaches now prioritise analytical understanding over commercial exploitation to address these exact pressures. Unsupervised learning methods might reduce certain biases by avoiding rigid categorisation structures (Caron et al., 2018). They do not eliminate ethical risks entirely, and questions of the representation, fairness, and interpretability persist when selecting a methodology. This study therefore adopts a cautious approach, with the focus entirely on exploratory analysis rather than deployment driven prediction. Explicitly acknowledging the social implications of applying machine learning to human data remains central to this work.

2.6 Critical Comparison of Existing Work and the Proposed Approach

Existing research shows that machine learning methods works well for analysing large fashion datasets from social media platforms. However, methodological gaps appear when comparing established methods with the proposed system. Most recent research relies on supervised learning where models use set labels and fixed fashion categories to organise their findings (Li and Zhou, 2024). While these frameworks provide interpretability and strong results in classification, they are limited by the assumptions built into their labels, which often prevents them from identifying trend that do not fit into pre-defined boxes. Fashion expression online is fluid, supervised models usually repeat what we already know instead of finding something new, or rather, they miss the subtle shifts that define the next fashion season. In contrast, this system focuses on discovery through unsupervised learning, which allows the data to form its own patterns without the use of pre-set categories, which fits with the idea that early trends are often quiet and spread out across different groups (Kalyanam and Quezada, 2021). By using multimodal embeddings and clustering methods, the system identifies hidden clusters that match what people post online. The choice between these methods often originates from the trade-off between how easily one can measure success and how much the system can adapt. Supervised systems are simple to test, however, they require people to constantly update those labels as styles change, which makes them hard to scale when things move quickly. In contrast, this

approach is more flexible as unsupervised methods adjust to new data on their own without needing constant human intervention. This makes them a better fit for the high-speed world of social platforms (where trends can die in a week), though this freedom brings its own problems. Without labels, it is hard to say for sure if the patterns the model finds are valid. And even if we do not use fixed categories, bias and uneven data still exist in social media sets (Binns and Veale, 2023). W These issues are addressed by using specific evaluation strategies. overall, the two methods work best together. One offers a solid structure while the other allows for discovery, and this work builds a framework that uses the best parts of both whilst being honest about their flaws.

2.7 Summary and Identified Research Gap

This literature review has examined fashion trend analysis within social media environments, focusing specifically on how machine learning processes large scale, multimodal data. Platforms such as Instagram and Pinterest provide rich visual and textual information. Combining image features with metadata improves the identification of stylistic patterns (Liu and Luo, 2020; Said and Alharbi, 2025). However, a critical evaluation of current methods reveals that much of this work remains tied to supervised learning frameworks. These models depend entirely on labelled datasets and predefined fashion taxonomies. They offer interpretability. But they also impose strict structural constraints. Models built this way lack sensitivity to emerging, unconventional, or weakly expressed trends, which is a major limitation given that social media fashion expression is highly fluid and culturally driven. Many existing systems demonstrate strong descriptive capability but struggle to capture early-stage trend formation or adapt to rapidly shifting data distributions. Recent advances in unsupervised and representation learning have introduced more flexible alternatives. These methods allow algorithms to identify patterns directly from data without relying on explicit labels (Caron et al., 2018). Such techniques align well with the exploratory nature of trend discovery. However, their application within fashion-specific contexts remains relatively underdeveloped. Systematic evaluation presents difficulties, a clear tension therefore emerges across the literature between flexibility and evaluability. Unsupervised approaches support the discovery of novel patterns, yet they introduce major challenges in interpretability and empirical validation because reliable reference labels do not exist. Algorithmic bias and data privacy further

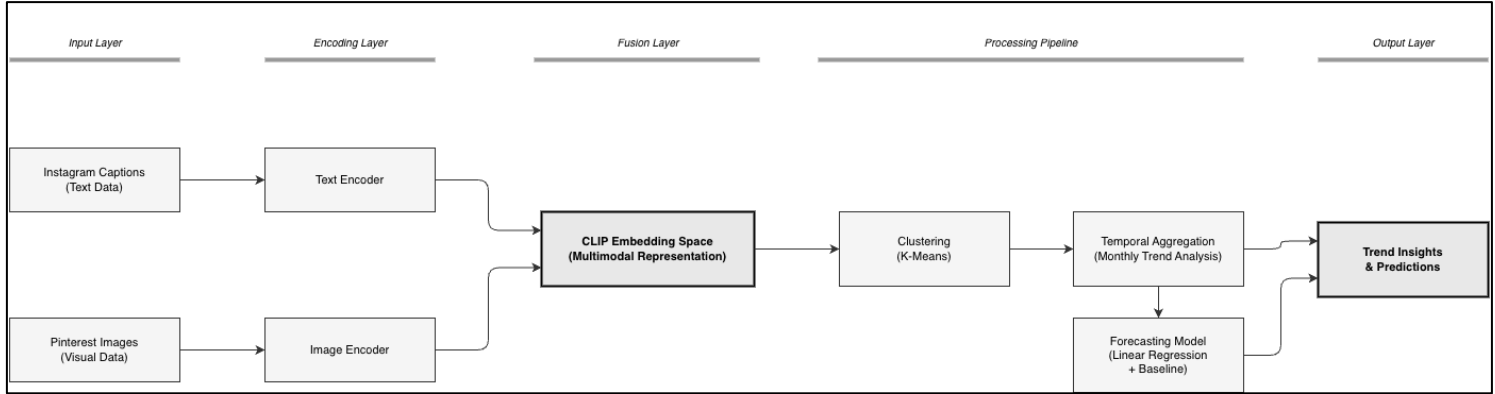
complicate the use of social media data (Binns and Veale, 2023). How can researchers balance this need for flexibility with rigorous validation? The findings above identify a clear research gap at the intersection of multimodal social media data, fashion analysis, and exploratory machine learning. The field lacks research integrating unsupervised, multimodal representation learning with explicit temporal analysis and structured evaluation of forecasting performance in fashion contexts. This study addresses that gap. This project proposes a new analytical framework, It combines vision-language embeddings, unsupervised clustering, and trend forecasting techniques to process Instagram and Pinterest data. By incorporating both exploratory pattern discovery and quantitative evaluation, this approach balances flexibility with analytical rigour, ultimately yielding a highly adaptable method for understanding fashion trends in complex environments.

3 Methodology

3.1 Research Design and Methodological Framework

The aim of this project is to investigate whether large-scale, multimodal social media data can reveal and predict fashion trends. Within this context, the challenge takes shape as the detection and temporal analysis of hidden fashion patterns within unstructured user-generated content, in contrast to standard supervised forecasting methods. Unlike conventional forecasting problems, where clearly defined labels and historical targets enable direct model training, fashion trends on platforms such as Instagram and Pinterest emerge organically through decentralised user behaviour. Such results produce datasets that prove multidimensional, cluttered, and loosely organized, often lacking reliable or consistent labelling. As a result, this problem is addressed through this investigative machine learning project, where the goal is to uncover meaningful structure within the data before prediction. This approach encourages the use of unsupervised learning techniques, enabling latent patterns to be identified without using predefined categories that overlook the evolving nature of fashion (Bishop, 2006; Aggarwal and Zhai, 2012). The overall system architecture designed to address this is illustrated in Figure 3.1.

Figure 3.1 Overview of the Proposed Trend Analysis Pipeline



This study implements this approach through a modular, data driven pipeline that combines multimodal representation learning with clustering, temporal analysis and forecasting, as shown in Figure 3.1. The heart of the system is the conversion of heterogeneous data into a common feature space, text from Instagram captions and Images from Pinterest are encoded using a vision-language model, which allows for this to happen. This design allows for comparisons across modalities and reveals underlying patterns, addressing the challenge of analysing datasets where modalities are not already paired. Following completing the representation learning phase, clustering methods are applied to group smaller data points into hidden structures, where in this study these are called fashion content archetypes. The resulting clusters function as adaptable trend spaces, unlike strict classification, allows the system to capture evolving patterns within the data. Within Figure 3.1, temporal aggregation is then used to track the popularity of these clusters over time, providing a base for identifying emerging fashion trends. The final forecasting stage models past cluster paths to forecast a brief term trend dynamic, which maintains focus on straightforwardness and clarity, which is consistent with standard forecasting methods (Hyndman and Athanasopoulos, 2018). Overall, the framework operates as a step-by-step pipeline, where each step targets a distinct element of the issue at hand. Thus, facilitating both initial exploration pattern discovery and structured evaluation within a complex and rapidly changing social media environment.

3.2 Data Collection and Dataset Construction

The analysis in this study derives from two complementary datasets from social media platforms. These datasets were chosen deliberately to capture the multimodal aspects of modern fashion styles. Instagram captions deliver a textual lens within the fashion

environment, to reveal how users describe outfits, styles, and aesthetic preferences through natural language. Meanwhile, Pinterest images supply a visual perspective, reflecting curated fashion content that emphasises stylistic composition, colour, and garment design. The use of publicly accessible data ensures transparency and reproducibility, while also avoiding the need for authenticated scraping or restricted data access (Zhang, Chen and Lee, 2019). Together, these sources enable the study to capture both descriptive and visual signals, aligning with the multimodal framework introduced in section 3.1 and illustrated in Figure 3.1. Table 3.1 summarises the datasets used in this study, including their key characteristics, scale, and limitations.

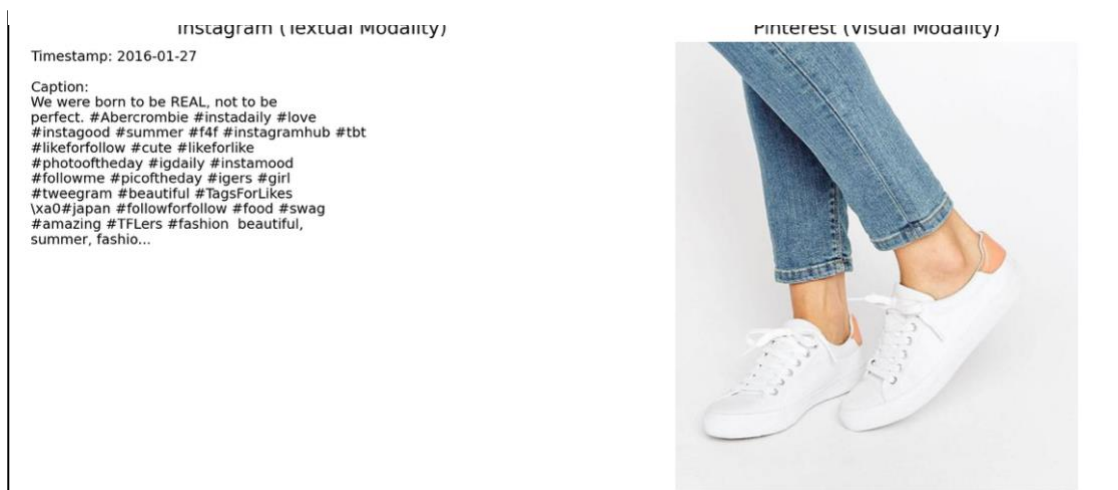
Table 3.1 Summary of Datasets Used in the Study

Dataset	Source	Data Type	Size	Key Features	Limitations
Shop The Look	Instagram	Text (captions)	24752	Hashtags, Descriptions, User language	No images, Noisy text
Pinterest-Fashion-Dataset	Pinterest	Images	1000	Visual Fashion Content	Recommendation Bias

As shown in Table 3.1, the Instagram dataset consists of a larger collection of caption data linked to fashion-oriented posts. Textual data is used as a substitute for visual trends, due to the limitations in image accessibility. The captions include hashtags, descriptive phrases, and informal language, which together provide insight into how users categorise and communicate fashion trends. However, this data is very noisy, as it features promotional content, irregular formatting, that differ widely. In contrast, the Pinterest dataset draws images suggested by platform recommendations, resulting in visually rich but potentially biased content. Unlike Instagram data, which reflects user-generated content directly, Pinterest data is influenced by platform recommendation algorithms that prioritise visually similar and popular (trendy) content. As a result, this dataset may reflect algorithmically curated trends rather than a fully representative distribution of user preferences, which introduces a source of structural bias that requires attention in later analyses. Pinterest-Fashion-Dataset is heavily influenced by Pinterest’s recommendation system, as it tends to emphasize visual themes or trends, thereby reducing the range of styles present in the dataset.

An important characteristic of the dataset is the imbalance between platforms, both in scale and in representational diversity. As shown in Table 3.1, the dataset is heavily skewed towards Instagram, with 24,752 textual entries compared to 1,000 Pinterest images. This imbalance introduces a structural bias, whereby textual patterns may be captured in greater detail than visual ones, potentially influencing downstream clustering and trend identification. Preliminary inspection further suggests that Pinterest content is relatively homogeneous in composition, often centred around similar product-focused imagery, which may limit the number of distinct patterns that can be identified through unsupervised methods. More broadly, fashion trends observed within social media are non-stationary and shaped by external influences such as seasonality, cultural events, and influencer activity, meaning the dataset represents a dynamic and evolving signal rather than a stable distribution. The diversity across data types, combined with the effects of these platform-specific algorithms, imposes a key limitation that is considered when interpreting the results. An illustrative comparison of the two data sources is shown in Figure 3.2, which highlights the distinction between textual and visual modalities used within the analysis.

Figure 3.2 Example instances from Instagram and Pinterest, illustrating the distinction between textual and visual modalities



3.3 Data Preprocessing and Feature Engineering

Social media data exhibits a heterogeneous and unstructured nature which requires a carefully designed preprocessing strategy. This ensures that textual and visual elements alike convert into forms appropriate for the following analytical tasks.

Fashion content is inherently noisy, informal, and context dependent, this preprocessing method emphasises preserving the semantic depth whilst applying precise cleaning steps to improve data quality. Preprocessing for the Instagram dataset centres on standardising captions and managing noise, Table 3.2 depicts how raw captions often contain informal language, emojis, excessive punctuation, and dense hashtag usage.

Table 3.2 Raw Caption Example

ImgURL	CreationTime	Caption	Hashtags
https://scontent.cdninstagram.com/t51.2885-15/...	1453883211	We were born to be REAL, not to be perfect. ...	beautiful, summer, fashion, love, cute, food, ...
https://scontent.cdninstagram.com/t51.2885-15/...	1453885002	Gotta run, but first, let me take selfie. Me...	teen, model, brunette, selfie, hollister, snap...
https://scontent.cdninstagram.com/t51.2885-15/...	1453862683	dress to impress\U0001f457	alexandani, cute, llbean, beanboots, anthropol...

Text is normalised through lowercasing and the elimination of irregular characters and encoding artefacts, while preserving hashtags due to their role as efficient labels for fashion styles, settings and categories in fashion. This selective cleaning ensures that meaningful semantic signals are retained, whilst reducing noise that could negatively affect representation learning. This changes from conventional natural language processing workflows, which often entail rigorous stopword elimination or manual feature engineering. Here, those steps yield to preserve the organic linguistic patterns that contemporary embedding models anticipate. Beyond textual disruptions, the dataset harbours structural irregularities like malformed entries and low information content. These are addressed through filtering steps that remove invalid records and reduce the influence of extreme outliers. Yet promotional or redundant material persists to some extent, which mirrors authentic social media patterns. However, this balance is important, as removing such content would risk distorting the underlying data distribution. For the Pinterest dataset, preprocessing focuses on image validation and accessibility. The verification of image URLs guarantees inclusion solely of accessible, valid visuals, which prevents downstream errors during embedding generation. In contrast to standard computer vision approaches, no handcrafted visual features are extracted. Instead, raw images are passed directly to the embedding model, which applies its own internal preprocessing, including resizing and

normalisation. This approach reflects a shift towards end-to-end representation learning, where feature extraction is performed implicitly by the model.

Aligning structures across data types stands as a pivotal preprocessing feature. Captions from Instagram and images from Pinterest stem from different formats, however both are transformed into a unified scheme consisting of a platform identifier, content field, and temporal metadata. As shown in table 3.3, both datasets are mapped into a consistent structure, enabling them to be combined into a single multimodal dataset. Table 3.4 shows this unified dataset and confirms successful integration and highlights the imbalance between platforms, with Instagram contributing a significantly larger portion of samples.

Table 3.3 Schema Mapping for Instagram and Pinterest (Feature engineering)

	platform	Image_path	timestamp	text
0	instagram	https://scontent.cdninstagram.com/t51.2885-15/...	2016-01-27 08:26:51	We were born to be REAL, not to be perfect. ...
1	instagram	https://scontent.cdninstagram.com/t51.2885-15/...	2016-01-27 08:56:42	Gotta run, but first, let me take selfie. Me...
2	instagram	https://scontent.cdninstagram.com/t51.2885-15/...	2016-01-27 02:44:43	dress to impress\U0001f457 alexandani, cute...
0	pinterest	http://i.pinimg.com/400x/88/70/b0/8870b0d38413...	NaT	These are white canvas lace-up sneakers with ...
1	pinterest	http://i.pinimg.com/400x/c3/1d/1e/c31d1e4b326c...	NaT	These are women's wedge sandals with a black ...
2	pinterest	http://i.pinimg.com/400x/48/a5/ab/48a5abadc1ce...	NaT	A pair of checkerboard slip-on shoes from Van...

Table 3.4 Unified Dataset Construction

Metric	Value
Dataset Shape	(25,752 , 4)
Platform	Count
instagram	24752
pinterest	1000

Data quality is further assessed through missing value analysis. As shown in Table 3.5, missing values are minimal across key fields, with only a small proportion found in the timestamp variable. These findings affirm that the preprocessing pipeline has successfully preserved essential information while ensuring dataset completeness for later analysis.

Table 3.5 Missing Values (Data Quality Check)

timestamp	0.038832
platform	0.000000
Image_path	0.000000
text	0.000000
dtype	float64

Temporal feature engineering is also performed to enable trend analysis. Timestamps are converted to a standard format, then merge into monthly periods, as shown in Table 3.6. This transformation provides a consistent temporal resolution for analysing trend evolution over time. During this process, anomalies in timestamps surfaced and was then addressed. Invalid dates such as 1970 due to encoding issues were removed. Table 3.7 displays how this sharpens the time span into a credible, uniform analytical frame.

Table 3.6 Timestamp Processing (Temporal Feature Engineering)

	CreationTime	year_month
0	2016-01-27 08:26:51	2016-01
1	2016-01-27 08:56:42	2016-01
2	2016-01-27 02:44:43	2016-01
3	2016-01-27 02:44:43	2016-01
4	2016-01-27 06:04:30	2016-01

Table 3.7 Filtering Invalid Dates (Noise Handling)

Stage	Minimum Date	Maximum Date
Before Filtering	1970-01	2016-01
After Filtering	2015-06	2016-01

Within this study, feature creation leans heavily on the embedding model's capabilities over human-led efforts. Traditional methods like TF-IDF give ways to direct semantic encoding via CLIP. CLIP encodes both text and images into a common embedding space, this setup allows cross modal comparison and pattern detection without requiring explicit pairing between captions and images. This allows the model to capture high-level semantic relationships across modalities, which forms the basis for clustering and trend identification in following stages of the pipeline.

3.4 Feature Representation (CLIP)

After the preprocessing stage, the next stage transforms both textual and visual elements from the data into a unified numerical format, which is suitable for machine learning analysis. Traditional approaches to feature representation often treat text and

images separately, relying on methods such as Term Frequency-Inverse (TF-IDF) for text and handcrafted visual elements for images (Salton and Buckley, 1988; Wang et al., 2015).

However, these strategies are limited in their ability to capture high level semantic relationships across modalities, particularly in the context of social media data, which is inherently multimodal and weakly labelled. This study counters shortcomings by employing Contrastive Language Image Pretraining (CLIP) as the primary mechanism for feature representations. CLIP functions as a vision-language model, one trained across large scale image-text pairs, enabling it to learn a shared embedding space in which semantically related images and texts are positioned closely together (Radford et al., 2021). In contrast to older workflows, which demand distinct handling and subsequent alignment of each modality, CLIP encodes captions alongside images to compact vector forms of fixed dimension. This aids for seamless analysis in later stages. Several elements underscore the appropriateness of CLIP within this investigation. Social media datasets rarely include formal annotations, which complicated the deployment of supervised models. CLIP navigates this challenge through its use of inherent language cues as guidance; it derives semantic alignments straight from imperfect, everyday pairing of images and text (Radford et al., 2021; Saharia et al., 2021). That method resonates with the nature of data from platforms like Instagram and Pinterest, where accompanying text links to visuals in informal ways, absent any rigorous tagging. Beyond that, fashion material thrives on context and visual nuance. It demands encodings that encapsulate intangible aspects, from stylistic choices to aesthetic vibes and cultural undertones. Evidence demonstrates that CLIP yields versatile, deeply semantic embeddings, rendering it ideal for pursuits in visual interpretation and reasoning across modalities. (Radford et al., 2021).

For implementation, every Instagram caption and each Pinterest image feeds into the CLIP encoder. The result consists of dense vector embeddings situated within one cohesive space, those embeddings then fuel the following clustering and examinations of trends. Working from this unified space allows straightforward contrasts between verbal descriptions and pictorial elements. In turn, such comparisons reveal unified fashion patterns that bridge the two modalities. This approach is consistent with recent work in social media fashion analysis, where multimodal learning has been shown to significantly improve the identification of emerging trends and patterns (Liu and Luo,

2020; Li and Zhou, 2024). An additional advantage of using CLIP is the reduced reliance on manual feature engineering. Rather than explicitly constructing features, semantic structure is learned implicitly within the embedding space, reflecting a broader shift in machine learning towards representation learning (Goodfellow, Bengio and Courville, 2016). This is highly beneficial in dynamic domains such as fashion, where predefined features may quickly become outdated or fail to capture evolving styles. Furthermore, the use of embedding based representations supports unsupervised learning techniques, such as clustering, which are well suited for exploratory analysis in large, unlabelled datasets (Caron et al., 2018).

Overall, the chosen strategy for features harness CLIP to connect textual and visual data effectively. It supports scalable scrutiny of fashion dynamics on social platforms, grounded in semantic depth. By embedding heterogeneous data into a common space, the approach lays a solid base for uncovering hidden patterns and shifts over time within the dataset.

3.5 Clustering Methodology (unsupervised)

Clustering acts as the core analytical component for this project, which uncovers hidden fashion archetypes within high-dimensional multimodal embeddings. Since no labelled data was available, the analysis employed an unsupervised learning method, specifically called the K-means clustering algorithm. This algorithm divides the data into k clusters whilst minimising the within-cluster sum of squares, as defined in Eq. (3.5.1): *K-Means objective function (within-cluster sum of squares)*.

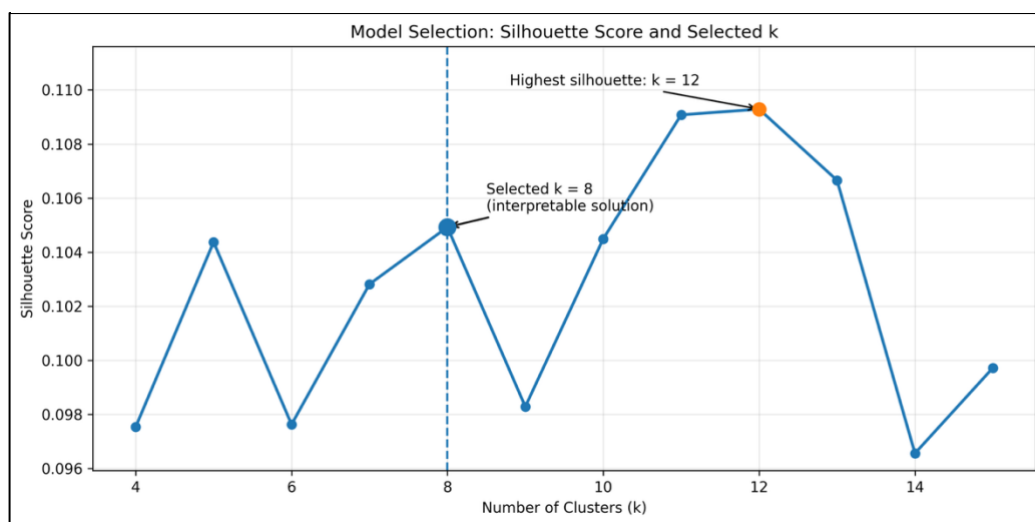
$$J = \sum_{i=1}^k \sum_{x_j \in C_i} \|x_j - \mu_i\|^2 \quad (3.5.1)$$

Where C_i represents cluster i and μ_i its centroid. The objective makes sure that data points in each cluster remains as similar as possible within the embedding space (MacQueen, 1967; Bishop, 2006). This algorithm proves it is suitable for the task, given its efficiency and effectiveness in high-dimensional vector spaces, including those generated by CLIP embeddings. The clustering is applied to joint embeddings that the CLIP model produces. In this setup, representations of Instagram captions and Pinterest images occupy a common semantic space. This approach supports grouping according to the conceptual similarity, that is independent from modality, allowing visually and textually similar fashion content to be clustered together. This

embedding-based clustering aligns well with modern approaches to unsupervised representation learning (Caron et al., 2018; Radford et al., 2021).

Determining the appropriate number of clusters (k), acted as a crucial step within the methodology. Silhouette scores were calculated over a series of potential values to assess the quality of each cluster. Figure 3.3: illustrates that the silhouette score reaches its maximum as $k=12$, suggesting the optimal separation based solely on quantitative measures. However, upon closer examination of the resulting clusters, higher k values appeared to cause fragmentation and diminished interpretability, with some clusters capturing overly narrow or platform specific patterns.

Figure 3.3 K-Selection Plot



Therefore, the value $k=8$ was chosen for the final model setup. Figure 3.3 illustrates this decision as well, highlighting the selected k next to the statistically optimal value. Although it does not achieve the maximum silhouette score, $k=8$ yields a more coherent and interpretable structure. In this way, the clusters capture meaningful fashion categories instead of noise or artifacts. Such an approach embodies a typical trade-off in unsupervised learning, balancing statistical optimisation against semantic interpretability (Aggarwal and Zhai, 2012). Figure 3.4 presents the resulting cluster distribution, illustrating the number of samples assigned to each cluster. The distribution exhibits moderate imbalance, as some clusters include a greater proportion of samples than others. This imbalance roots from the inherent frequency of dominant fashion styles within social media rather than a limitation in the clustering process. Larger clusters align with broad fashion styles, whereas smaller ones capture niche or emerging fashion categories.

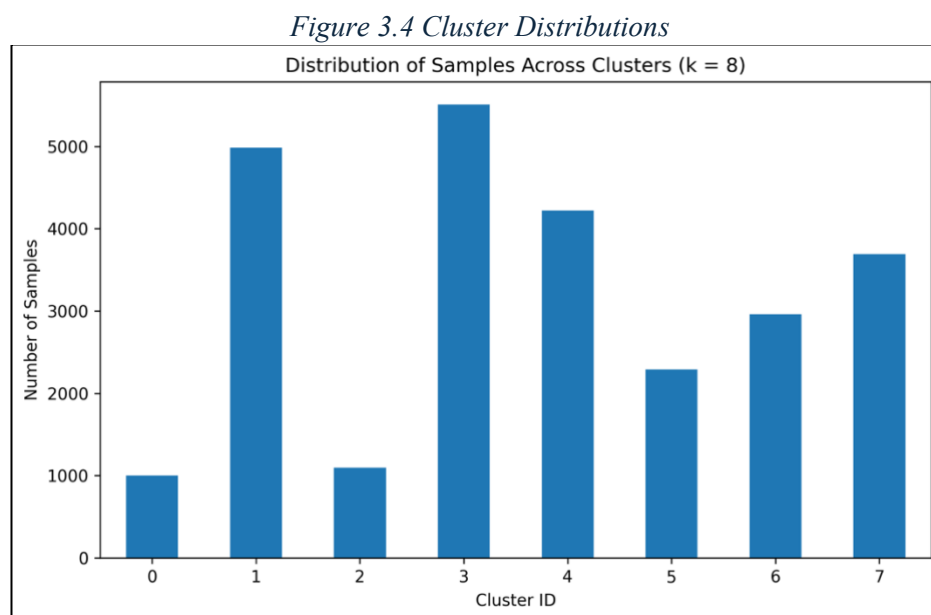


Table 3.8 illustrates the analysis of exemplar samples from the selected clusters. These samples reveal the semantic traits that define each group, one cluster for example, resolves around promotional fashion posts that include brand references and detailed outfit breakdowns. Another focused on menswear tailoring, with repetitive hashtags like “#menswear, #dapper”. This demonstrates that the clustering process successfully groups content based on meaningful stylistic and contextual features. Pinterest’s dataset adds less variety than Instagram’s, since many images shared similar visual elements caused by the recommendation system. Such patterns shape

the clusters, yet they remain intact on purpose, as they imitate genuine platform dynamics rather than artificial alterations (Binns and Veale, 2023).

Table 3.8 Instagram Cluster Exemplar

Cluster	Label	Example
0	Commercial Fashion Promotions	Happy Monday 🙌 Outfit of the day featuring Zara blouse and Sandro boots. #ootd #style
5	Menswear Tailoring & Formal Style	#mensfashion #dapper #suitandtie #menswear #style #wiwt

Overall, the clustering methodology shows how unsupervised learning on multimodal embeddings can reveal interpretable fashion archetypes. This approach integrates quantitative evaluation, model choices guided by domain knowledge, and qualitative validation through exemplars. As a result, the approach establishes a solid base for later trend analysis and forecasting.

3.6 Trend Analysis

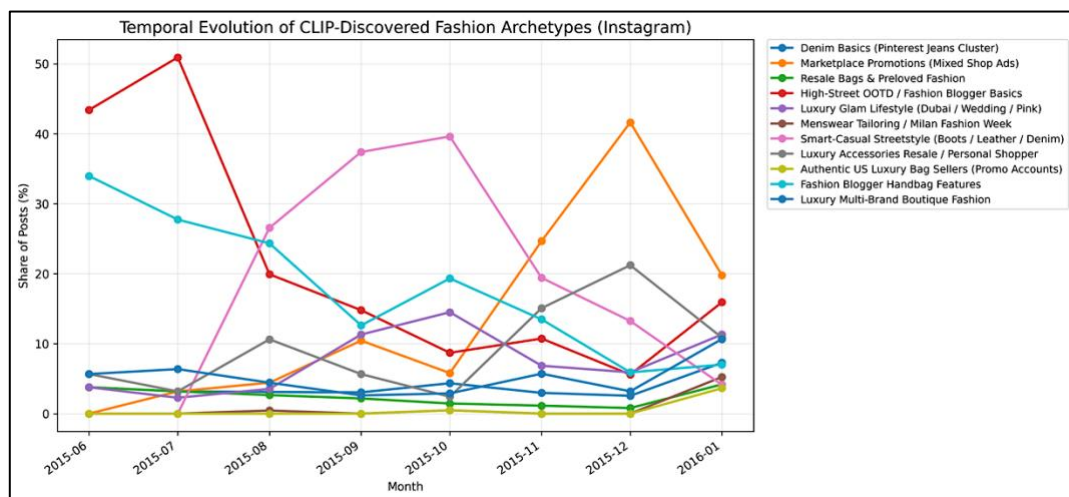
After clustering, a temporal analysis examines the evolution of identified fashion trends over time. The goal is to convert static cluster assignments into interpretable time-series signals, which helps the detection of emerging and declining trends in social media data. This is achieved by collecting data monthly, normalising the cluster frequencies, and calculating trend momentum. The first step involves gathering cluster assignments into monthly intervals. Each post links to a timestamp that converts to a standardised monthly format, these monthly cluster counts are then calculated as percentage shared to adjust to changes in overall posting volume. Table 3.9 illustrates how this process transforms raw counts into a consistent format, with each row totalling 100% for reliable temporal comparisons. Such normalisation guarantees that changes capture shifts in relative popularity rather than fluctuations in data volume (Hyndman and Athanasopoulos, 2018).

Table 3.9 Monthly percentage share of posts assigned to each cluster.

Month	0	1	2	3	4	5	6	7
2015-06	0.0	0.0	1.9	49.1	11.3	1.9	30.2	5.7
2015-07	0.0	3.6	0.5	56.8	7.3	0.0	26.4	5.5
2015-08	0.0	4.4	0.4	27.4	7.1	23.9	23.9	12.8

Building on this, the temporal dynamics are analysed by tracking the share of each cluster over consecutive months. Figure 3.5 illustrates the progression of these cluster proportions through time, which uncovers varied behavioural patterns across fashion archetypes. Some clusters display sharp peaks followed by decline, which is a sign of short-lived trends. Others, by contrast, reveal steady increases or persistent stability, that point to enduring stylistic elements. Clusters linked to Promotional Material, for instance, show high volatility, whereas those ties to Menswear Tailoring or Luxury Fashion display more consistent trajectories. Such observations highlight the heterogeneous and dynamic nature of fashion trends within social media environments (Kalyanam and Quezada, 2021).

Figure 3.5 Cluster Trend Evaluation Over Time

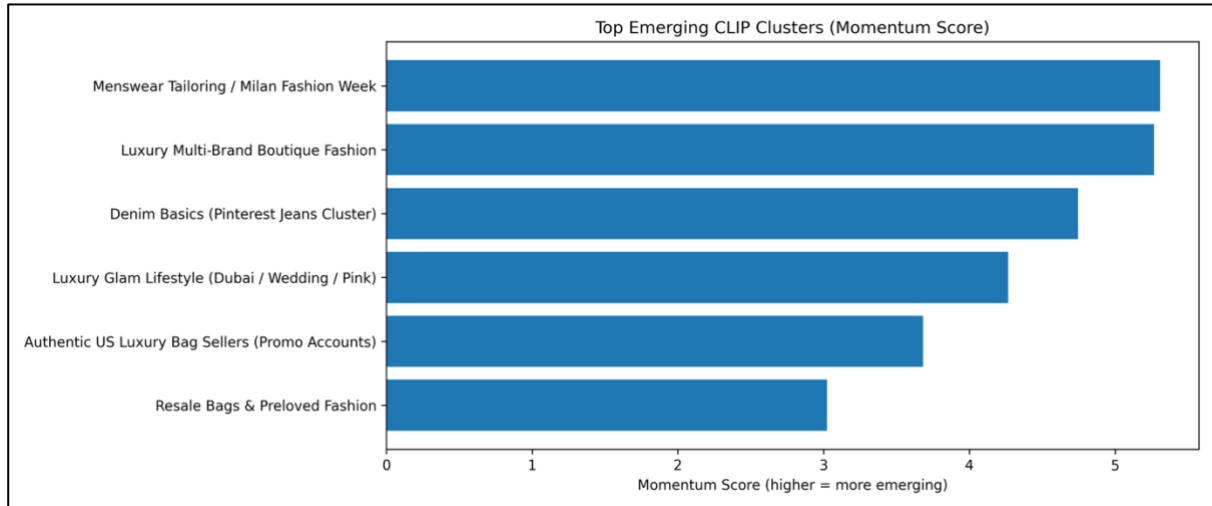


To define a “Fashion Trend”, this study introduces momentum-based measures which are derived from changes in cluster share over time. This method prioritises the rate of increase in representation over the popularity. As Figure 3.6 demonstrates, clusters receive a ranking based on their momentum score, where higher values indicate stronger upward trajectories. This enables the identification of emerging trends that may not yet dominate in volume but demonstrate rapid growth. For example, clusters tied to Menswear Tailoring and Luxury Boutique fashion, score high on momentum, which suggests their rising influence over the study period. This strategy correlates with previous studies in social media trend analysis, in which temporal shifts and growth patterns serve as key indicators of emerging trends and behaviours (Li and Zhou, 2024; Said and Alharbi, 2025). By emphasising relative changes instead of fixed

frequencies, this method detects initial signs of trend emergence, which is a critical advantage within the fashion field.

The overall temporal modelling framework provides a structures method for analysing trend evolution. By collecting data monthly, it maintains consistency across periods,

Figure 3.6 Emerging CLIP Clusters (Momentum Score)



while cluster share analysis uncovers the core patterns, and momentum scoring delivers a rigorous way to identify trends that are gaining traction. These elements together lay the groundwork for the following forecasting stage, in which time series models assess the degree to which trends can be forecasted reliably.

3.7 Forecasting Approach

This study evaluates the predictability of fashion trends through a simple time-series forecasting approach. This method applies to the monthly distributions of cluster shares obtained from Section 3.6. Given the short time frame and substantial fluctuations in social media data necessitate a focus on models that prioritise interpretability and simplicity over intricate designs. Linear regression serves as the main forecasting tool, it projects a cluster's future share by drawing on its past performance. More precisely, for every cluster, the model assumes a linear connection between time and the cluster's share, by the Eq. (3.7.1): *Linear Trend Model*.

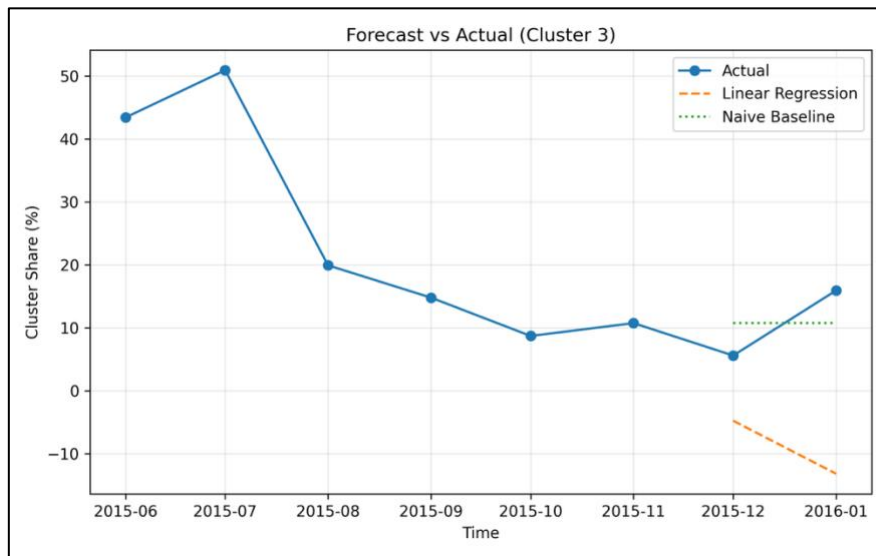
$$y_t = \beta_0 + \beta_1 t + \epsilon_t \quad (3.7.1)$$

Where y_t represents the cluster share at time t , β_0 and β_1 are parameters, and ϵ_t captures residual error. This approach is frequently used as a baseline for trend estimations because it is straightforward and easy to interpret (Hyndman and Athanasopoulos, 2018). To provide a meaningful comparison, a naïve baseline model is also implemented. This model assumes that the position that the optimal forecast for the upcoming time step consists simply of the latest observed value, shown in Eq. (3.7.2): *First-order Difference Equation* .

$$\hat{y}_{t+1} = y_t \quad (3.7.2)$$

In time series forecasting, the naïve model is particularly important, as it sets a basic performance benchmark that more elaborate models must outperform to demonstrate their worth. Figure 3.7.3 presents a representative example of forecasting performance for a single cluster. The actual cluster share shows significant fluctuation over time, marked with a steep drop and then some partial recovery. The linear regression model attempts to capture the overall downward trend, but forecasts this trajectory too aggressively, which results in unrealistic negative predictions. In contrast, the naïve baseline offers a stable estimate by simply propagating the last observed value. This example highlights a key limitation of linear models in high volatile environments, they manage to track overall trends yet often overlook abrupt shifts in the near term.

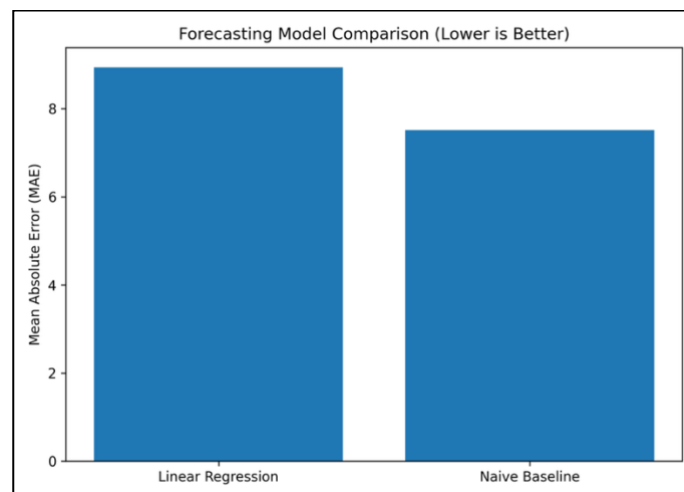
Figure 3.7 Forecast vs Actual for Cluster 3



Performance evaluation across all clusters relies on measuring forecasting accuracy with Mean Absolute Error (MAE). Figure 3.8 reveals that the naïve baseline yields a

lower MAE than linear regression does. Therefore, this suggests that the straightforward persistence-based method surpasses the trend-based model for the given dataset. Such results highlight the fundamental unpredictability of fashion trends on social media. These trends differ from structured economic or seasonal data, as fashion signals are influenced by rapidly changing factors such as platform specific patterns (Kalyanam and Quezada, 2021; Said and Alharbi, 2025). As a result, past trends often fail to offer a solid foundation for reliable forecasting. The decision to use simple models is intentional, because the limited time span of monthly data limits the effectiveness of more complex models such as ARIMA or neural networks, which demand longer sequences to generalise effectively. Secondly, the use of interpretable models allows for clearer evaluation of whether any predictive signal even exist, in addition a strong baseline facilitated meaningful evaluation for future improvements. Overall, the results suggest that while linear regression captures general directional trends effectively, it does not consistently outperform a naïve baseline. This reinforces the conclusion that short-term fashion trend forecasting remains a challenging task, where simple models offer a solid and clear basis for evaluation before exploring more advances techniques.

Figure 3.8 Forecasting Model Comparison



3.8 Evaluation Metrics

This study assesses the performance of forecasting by means of Mean Absolute error (MAE) and Mean Absolute Percentage Error (MAPE). These metrics represent standard metrics for time series prediction (Hyndman and Athanasopoulos, 2018). MAPE is defined in Eq.(3.8.1): *Mean Absolute Percentage Error (MAPE)*.

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right| \quad (3.8.1)$$

Where y_t represents the actual value and $\hat{y}_t$ represents the predicted value at time t . MAPE delivers a measure of error that is scale independent, making it appropriate for evaluating clusters that vary with different magnitudes of their shares. In addition, MAE complements MAPE by supplying a straightforward and interpretable measure of average absolute difference between predicted and observed values. Whereas MAPE proves more vulnerable to the effects of small denominators, MAE exhibits greater stability, particularly when the shares within clusters approach zero.

Importantly, the evaluation of model performance occurs in relation to a naïve baseline, rather than isolation. This comparison approach confirms that improvements in prediction stem from the model's true strengths, and not from peculiarities in the data. Section 3.7 illustrates that this naïve baseline continues to perform competitively, which highlights the difficulty of forecasting highly dynamic social media driven trends.

3.9 System Implementation

The system is implemented as a modular data processing pipeline, structured to promote scalability, reproducibility, and distinct task separation. The pipeline begins with data ingestion and preprocessing (Section 3.3), followed by multimodal feature using CLIP embeddings (Section 3.4). Those embeddings feed into the clustering phase in Section 3.5, where fashion archetypes are identified. Later steps handle temporal aggregation and trend analysis in Section 3.6, culminating in the application of forecasting models as outlined in Section 3.7. This implementation relies on Python uses libraries such as pandas for data manipulation (McKinney, 2010), NumPy for numerical computation, and scikit-learn for clustering and regression tasks (Pedregosa et al., 2011). For embedding generation, the CLIP model is used (Radford et al., 2021), facilitating a cohesive depiction of both text and images. To increase efficiency and to support iterative testing, intermediate results like embeddings and cluster assignments, are cached to improve efficiency and allows incremental experimentation. Lastly, an interactive dashboard was created through Streamlit, to visualise cluster distributions, the evolutions of temporal trends, and forecasting outputs. This dashboard functions as the final interface within the pipeline, as it allows for exploration and validation of the models results. The overall design supports a

reproducible process, permitting independent execution of stages or integration within a complete pipeline.

3.10 Ethical, Legal and Data Considerations

This study pulls public social media data on Instagram and Pinterest, to avoid any personally identifiable information, and the focus remains on collected content patterns rather than the individual users. This approach aligns with key data protection rules under the General data Protection Regulation (GDPR), by minimising privacy risks and avoiding intrusive data usage. In contrast, ethical considerations remain relevant, social media data gets shaped by platform algorithms and user behaviours, which introduce systematic biases. For example, certain demographics or aesthetic styles may be overrepresented due to influencer activity or recommendation systems (Binns and Veale, 2023). Additionally, Pinterest offers a good example, as it reflects a curated and algorithmically filtered environment, which may not fully represent broad fashion trend. These factors are acknowledged as potential sources of bias, and results are interpreted with this limitation in mind.

3.11 Limitations Within the Methodology

There are few limitations that arise from the methodological choices made in this study. Firstly, the reliance on Pinterest data introduces a degree of sampling bias, since the content reflects algorithmic recommendations rather than a neutral or random selection of fashion images. The clustering results illustrate this issue, as certain visual themes are disproportionately grouped, which reduced the overall diversity of those clusters. Secondly the approach is based on unsupervised learning, meaning that the clusters are not guided by predefined labels. This allows the identification of trends that are unexpected, however without ground truth labels makes it difficult to quantitatively validate the semantic correctness of clusters. Thirdly, the dataset covers only a limited timeframe, and with such a narrow temporal window affect the trend analysis and forecasting. Short time series reduces the reliability of statistical modelling and limits the ability to capture longer-term seasonal patterns. Finally, the use of simple models like linear regression and naïve baselines, restricts predictive performance. These choices prioritise interpretability and stability, yet they fail to capture complex, non-linear dynamics inherent in fashion trends. Section 3.7 demonstrates this, where even basic baselines remain competitive, highlighting the

broader challenge of forecasting in highly dynamic and noisy social media environments.

Collectively, these constraints suggest that while the methodology provides a structure and interpretable framework for analysing fashion trends, further enhancements such as more diverse datasets, extended timelines, and a sophisticated modelling could be more advanced.

4 Results

4.1 Clustering Results

In the clustering stage, the model generates a collection of latent fashion archetypes drawn from the multimodal embedding space outlined in Section 3.5. These results are presented in terms of model selection, cluster distribution, and qualitative interpretation of cluster content. The selection of the number of clusters (k) is guided by silhouette analysis over various candidate values. Figure 4.1 demonstrates that the silhouette score at $k=12$ is approximately 0.109, which suggests the best separation from a purely quantitative perspective. However, the final model adopts $k=8$ at 0.105, this setup yields more coherent and interpretable grouping of fashion content. Although the silhouette score is slightly lower, the decision minimises fragmentation while preventing the appearance of narrow definitions or redundant clusters. Such a balance represents a common compromise in unsupervised learning, weighing statistical precision against meaningful semantic structure.

Figure 4.1 Silhouette scores across candidate values of k , with selected $k = 8$.

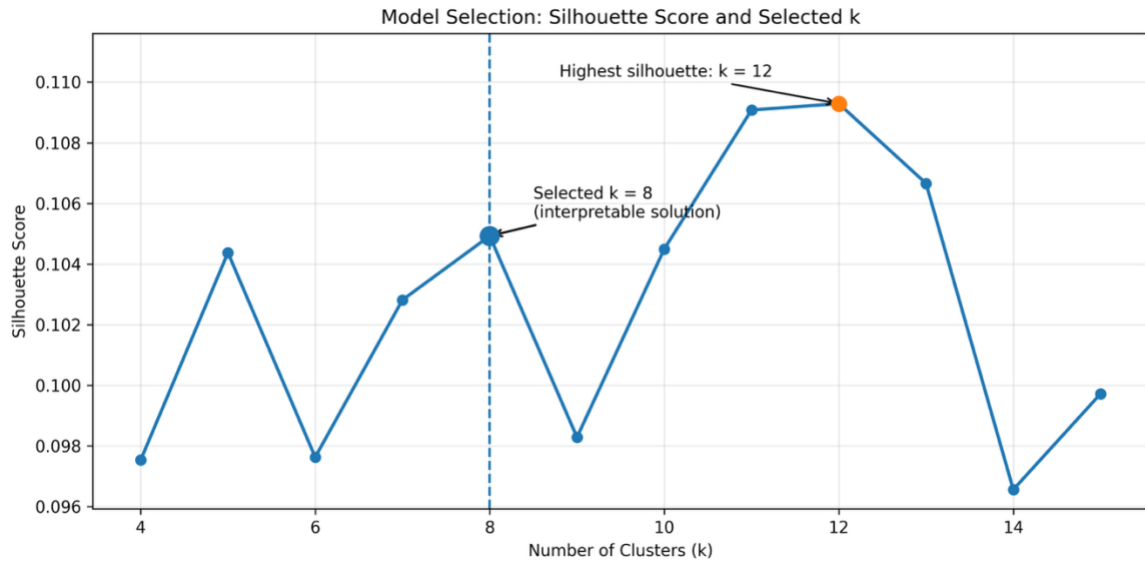


Figure 4.2 illustrates the resulting cluster distribution, which shows the number of samples assigned to each cluster. The distribution appears moderately imbalanced, as the largest cluster contains approximately 5,000 samples, and the smallest cluster contains around 1,000 samples. This imbalance arises from the inherent frequency of prominent fashion styles in the dataset, rather than a modelling artefact. The larger clusters align with more prevalent styles or content types, whereas the smaller ones represent specialised or less common patterns.

Figure 4.2 Distribution of samples across clusters ($k = 8$)

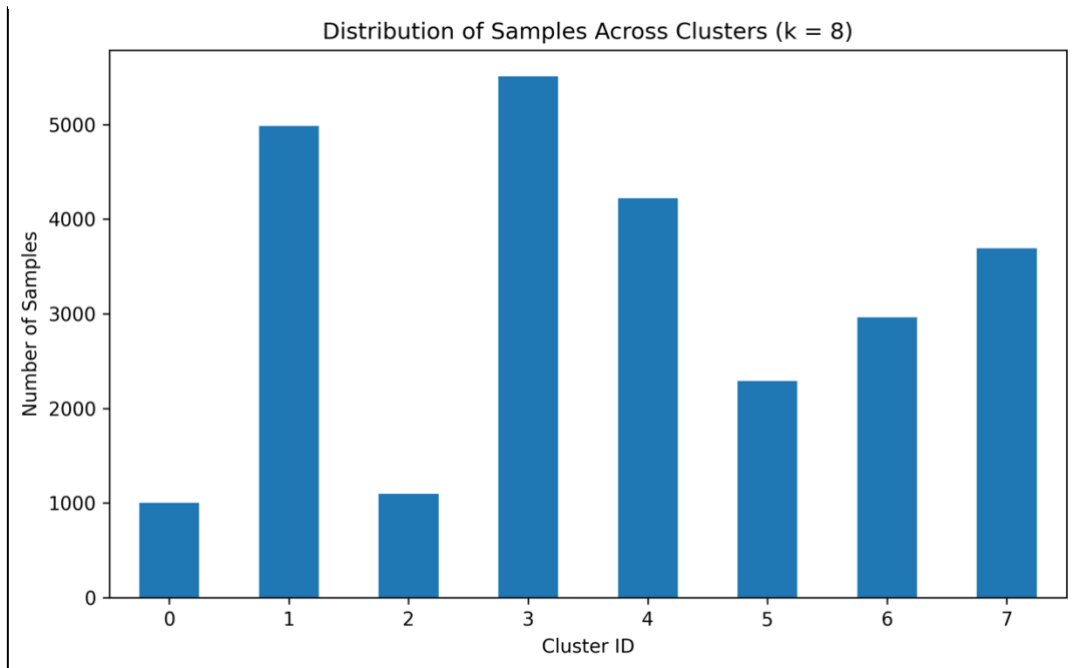


Table 4.1 presents representative examples that allow for an assessment of the clusters semantic coherence. These examples demonstrate that the clustering process successfully groups content based on meaningful stylistic and contextual features. One cluster, for example, canters on promotional fashion material, which features brand mentions alongside detailed outfit breakdowns. Another focuses on menswear tailoring, as evidenced by the repeated application of specialized hashtags like #menswear and #dapper. Such patterns indicate that the clustering corresponds closely to established fashion genres and the typical structures found in user-created social media posts.

Table 4.1 Representative examples of selected clusters.

Cluster	Label	Example
0	Commercial Fashion Promotions	Happy Monday 🙌 Outfit of the day featuring Zara blouse and Sandro boots. #ootd #style
5	Menswear Tailoring & Formal Style	#mensfashion #dapper #suitandtie #menswear #style #wiwt

The clustering results show that overall, as an unsupervised method, successfully divides the multimodal dataset into meaningful and interpretable fashion archetypes.

Quantitative model selection, along with distributional analysis and qualitative validation, demonstrates the resulting clusters offer a structured representation of fashion content, forming the basis for subsequent temporal trend analysis.

4.2 Temporal Trend Analysis

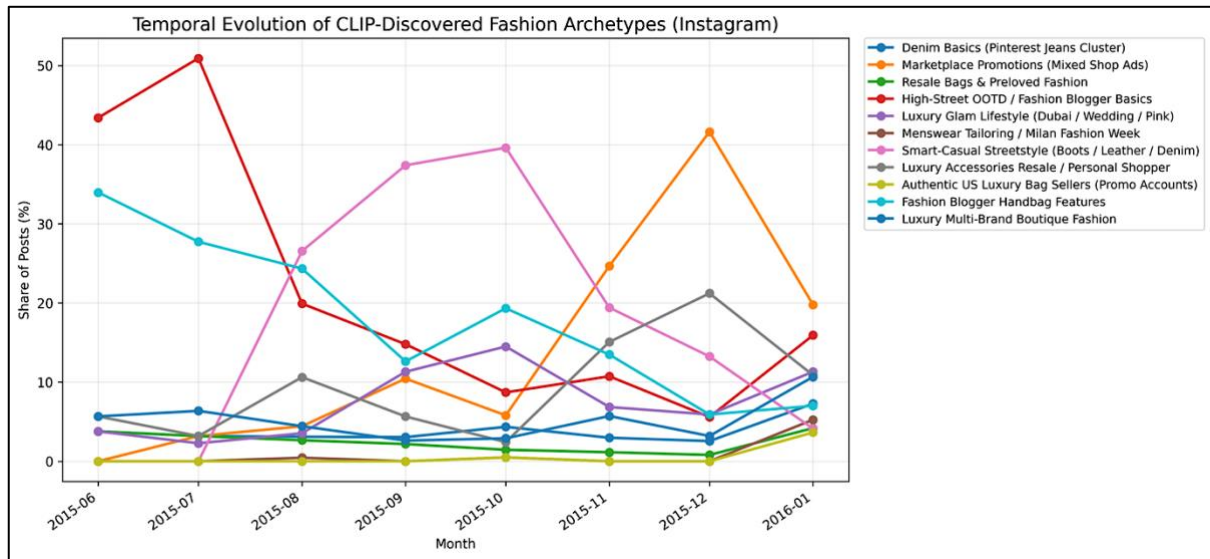
After identifying the clusters, temporal analysis is conducted to examine how the evolution of these fashion archetypes evolves over time. The findings appear in monthly cluster shares, overall trend evolution, and momentum-based ranking of emerging patterns. For a reliable comparison across periods, the analysis of the accumulates cluster frequencies monthly and normalises them as percentage shares. Figure 4.2. illustrates this, where each row captures the proportion of posts assigned to a specific cluster in that month, with all values totalling 100%. For example, Cluster 3 accounts for over 50% of posts in July 2015, while several clusters below 5% throughout the observed period. Such normalisation highlights relative shifts in cluster prominence rather than differences in overall data volume.

Table 4.2 Monthly percentage share of posts assigned to each cluster.

Month	0	1	2	3	4	5	6	7
2015-06	0.0	0.0	1.9	49.1	11.3	1.9	30.2	5.7
2015-07	0.0	3.6	0.5	56.8	7.3	0.0	26.4	5.5
2015-08	0.0	4.4	0.4	27.4	7.1	23.9	23.9	12.8

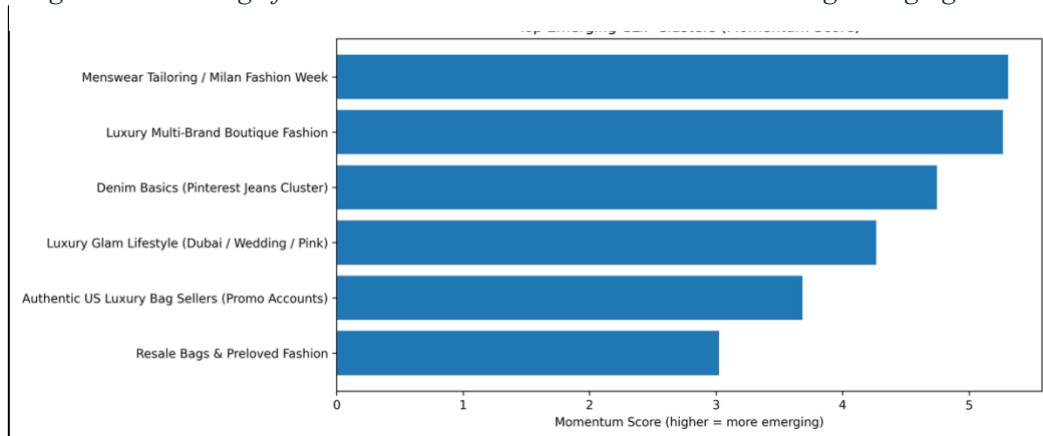
Figure 4.3. depicts the temporal evolution of cluster shares, tracing the behaviour of each cluster shares, which traces the behaviour of each cluster through successive months. The results indicate that cluster dynamics vary considerably over time. Certain clusters show noticeable increases in share, while others display gradual declines or periods of fluctuation. These patterns underscore the variability in how different types of fashion content appear across time.

Figure 4.3 Temporal evolution of cluster shares across monthly intervals.



To quantify shifts in cluster prominence, a momentum bases measure is applied. This approach captures the rate at which each clusters share changes over time. Figure 4.4 displays the resulting rankings, with clusters arranged by their momentum scores. The highest momentum score is observed for the top ranked cluster at approximately 5.3, with the following clusters showing progressively lower values. Higher scores signal robust upward movement in representation, whereas lower scores reflect slower growth or decline. The findings reveal that certain clusters achieve notably higher momentum scores compared to others, highlighting varying paces in how cluster shares develop across the study period.

Figure 4.4 Ranking of clusters based on momentum scores indicating emerging trends



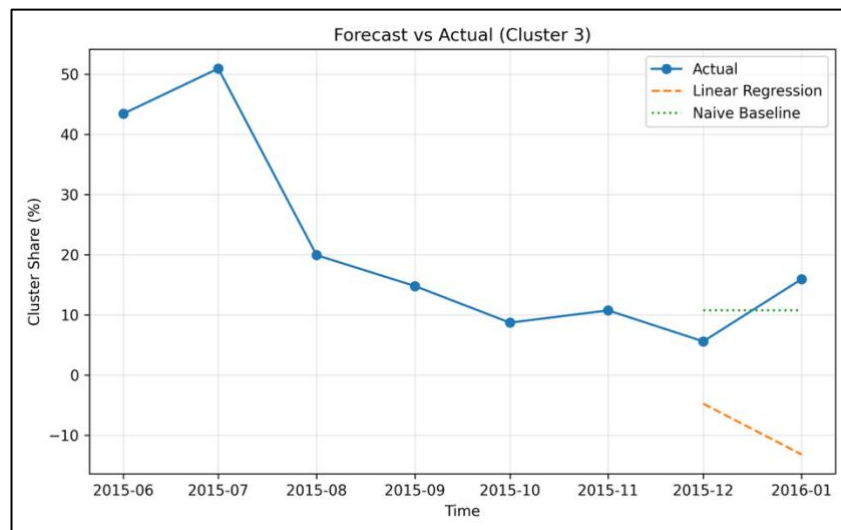
Overall, the temporal analysis demonstrates that cluster distributions are not static, but instead exhibit varying patterns of growth, decline, and fluctuation over time. These

results offer a clear framework for understanding the evolution of fashion related content across the dataset thus supporting further forecasting analysis.

4.3 Forecasting Results

To evaluate the predictability of the fashion trend identified earlier, forecasting models are fitted to the monthly time series of cluster shares obtained from Section 4.2. These results appear through an example of model behaviour and a comparison of forecasting accuracy across all clusters. Figure 4.5 provides a clear example of how forecasting operates, by juxtaposing the actual cluster share against forecasts from linear regression and the naïve baseline. The observed series exhibits noticeable variability, for instance, the cluster share decreases from above 50% to below 10% over the observed period. While the linear regression approach tracks the overall direction of the trend and yields a steady path drawn from past data points, the naïve baseline simply projects a flat line by repeating the last observed value. The differences between the predicted and actual values highlight variations in how each model responds to changes in the underlying time series.

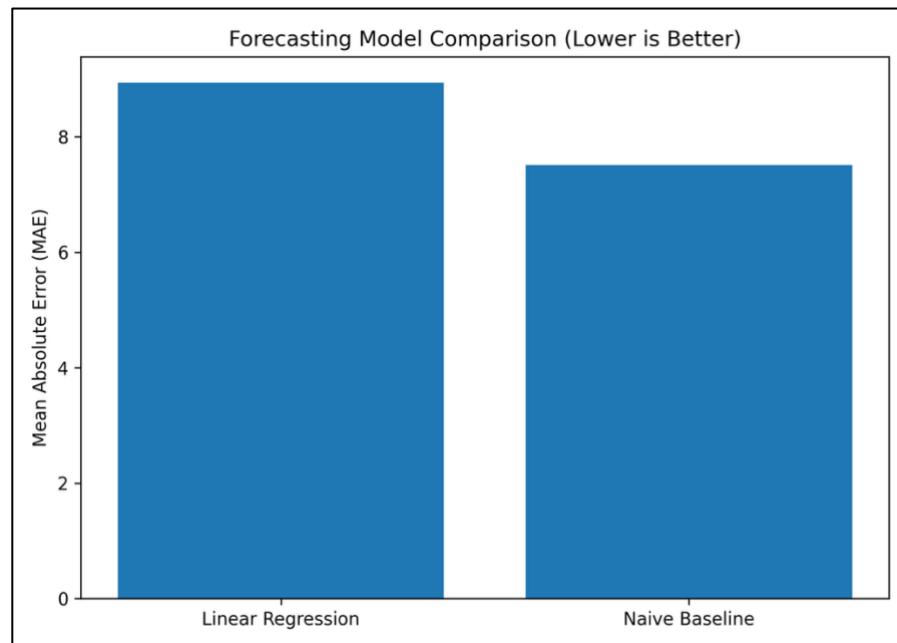
Figure 4.5 Forecast vs actual cluster share using linear regression and naïve baseline.



To evaluate performance on a broader scale, the accuracy of forecasting is assessed across all clusters through Mean Absolute Error (MAE). Figure 4.6 presents this comparison, the naïve baseline achieves an average MAE of 7.5, compared to around 9.0 for the linear regression model. It is evident that the naïve baseline yields a lower average MAE compared to the linear regression model. Moreover, across the dataset,

the naïve approach generates predictions that align more closely with observed values on average.

Figure 4.6 Comparison of forecasting performance between linear regression and naive baseline using mean absolute error (MAE).



The overall results from the forecasting indicate that both models can produce predictions drawn from historical cluster patterns, although they exhibit distinct levels of accuracy. This comparison reveals differences in predictive performance among the clusters and establishes a quantitative foundation for assessing the forecasting methods employed within this study.

4.4 Summary of Key Findings

In this study the results demonstrate that the proposed methodology effectively identifies and analyses fashion-related patterns in multimodal social media data. The clustering process divides the dataset into eight distinct groups, each embodying a coherent fashion content archetype, shaped by shared semantic traits. These clusters vary in size with the largest containing over 5,000 samples and the smallest 1,000. A temporal analysis uncovers diverse behavioural patterns across these clusters over time. Certain clusters sustain stable proportions. Others, however, display noticeable fluctuations, including periods of growth and decline. The momentum based ranking further highlights differences in the rate at which cluster shares evolve, indicating that

certain clusters experience stronger upward movement compared to others. The forecasting outcomes demonstrate that linear regression and naïve baseline models can produce predictions from historical cluster behaviour. This is reflected in the lower average MAE of approximately 7.5 for the naïve model compared to around 9.0 for linear regression. However, evaluating forecasting performance reveals that the naïve baseline yields lower average error rates across clusters, suggesting that straightforward persistent base forecasts are competitive in this dataset. In summary, these findings affirm that unstructured social media data yields meaningful fashion patterns with clear temporal dynamics. The forecasting results, meanwhile, expose differences in predictive precision. Such insights lay the basis for deeper exploration in the following discussion section.

5 Discussion and Analysis

5.1 Interpretation of Results

The result from this study provides clear evidence that multimodal social media data can reveal distinct fashion patterns, while also highlighting important limitations in how these patterns evolve over time and can be predicted. These insights unfold through an analysis of clustering outcomes, temporal shifts, and forecasting effectiveness. The clustering results reveal how the multimodal embedding method effectively organises fashion content into clusters that hold conceptual consistency. Distinct clusters such as Promotional Fashion Content, Menswear Tailoring, and Luxury Oriented styles suggests that the model can capture meaningful categories that reflect real-world fashion behaviour. This is supported by external fashion reports from highly influential fashion and lifestyle magazines, which indicate that key 2016 trends included structured tailoring, denim styles, and luxury aesthetics (Vogue, 2016a; Vogue, 2016b; Vogue Arabia, 2016g). The consistency between cluster trends and establishes trends affirms the value of the embedding and clustering approach for deriving broad fashion ideas from unstructured data. However, the uneven popularity of specific clusters, especially those tied to high street outfits and promotional items signals an imbalance in the distribution of fashion data on social media. These clusters claim a substantial volume of posts, implying that everyday commercial content occupies a significant share of the dataset. This reflects the nature of platforms such as Instagram, where influencer marketing and brand promotion play a central role in shaping visible fashion

content (Binns and Veale, 2023). Consequently, the clustering results do more than map trends, they also reflect platform specific dynamics, in which algorithms for engagement and recommendations influence visibility rather than purely organic trend formation.

A temporal analysis further reveals that fashion trends identified within the dataset are highly dynamic. Figure 4.3 illustrates substantial fluctuations over time, where some clusters experience rapid increases followed by sharp declines. Such patterns imply that social media fashion trends tend to be short-lived and subject to rapid changes in user behaviour. Although certain clusters maintain a degree of stability, others display instability, which demonstrates that trends in this domain rarely follow smooth or predictable trajectories. These observations align with earlier studies indicating that social media trends are influenced by factors such as virality, influencer activity, and platform specific algorithms (Kalyanam and Quezada, 2021; Said and Alharbi, 2025). The momentum bases ranking provides further insight into emerging trends, highlighting clusters that exhibit strong upward movement over time. The identification of Menswear Tailoring and Luxury related clusters as high momentum segments aligns with industry observations from 2016 where structured silhouetted and premium fashion gained popularity (Vogue, 2016a; Vogue UK, 2016c; Elle UK, 2016i). This suggests that the model can detect early signals of trend emergence, rather than simply identifying dominant categories. However, the analysis also reveals gaps in representing all real-world trends. Aspects unique to runways, including colour palettes, fabric details, and seasonal styling trends are less clearly represented in the clustering output, reflecting the limitations of both the dataset and the representation method. The forecasting outcomes expose key difficulties in forecasting fashion trends under these conditions. Figure 4.6 illustrates how the naïve baseline surpasses the linear regression model with respect to the mean error. This result points to the superior reliability of basic persistence forecasts over trend driven projections for the dataset. Linear regressions inability to exceed the baseline performance signals an absence of consistent linear patterns in the time series, which complicated modelling through straightforward parametric methods. Figure 4.5 further clarifies this issue, where linear predictions diverge significantly from actual values during periods of rapid change.

The challenges regarding forecasting rise from the inherent volatility and non-stationarity of social media data. Unlike traditional economic or seasonal datasets, fashion trends are influenced by unpredictable external factors, including cultural events, viral content, and shifts in user attention. Consequently, historical patterns often fail to serve as dependable guides for anticipating future progression. This observation corresponds with existing literature which emphasises the complexity of modelling social media trends due to their dynamic and rapidly evolving nature (Kalyanam and Quezada, 2021; Said and Alharbi, 2025). Overall, the results indicate that although the proposed methodology excels at identifying and examining fashion trends, it proves less effective when forecasting. The clustering and temporal analysis stages demonstrate strong alignment with real world fashion behaviour, particularly in capturing broad categories such as denim, menswear, and luxury fashion. However, the forecasting stage reveals fundamental limitations in predictability. These results highlight the strengths and constraints of data driven approaches to fashion trend analysis, providing a foundation for further explorations and methodological refinement.

5.2 Significance of Findings

The results from this study hold important implications for fashion trend analysis and the broader application of machine learning to social media data. These findings show that significant fashion patterns emerge from unstructured, multimodal content. The results highlight the growing value of data driven methods in an industry that is dependent on expert opinion and top-down forecasting. From an industry perspective, the ability to identify fashion trends directly from social media delivers a significant advantage in terms of speed and responsiveness. As traditional trend forecasting depends on runway analysis, seasonal reports, and editorial curation, all of which introduce delays between trend emergence and recognition. In contrast, this study reveals how social media data tracks trends as they unfold in real time, momentum-based measures enable the detection of emerging trends before they become dominant, as seen in clusters like Menswear Tailoring and Luxury Fashion. This suggests that platforms such as these serve as early signals for fashion shifts, giving designers, retailers, and marketing teams enough time to respond quickly to changing consumer preferences. A key contribution of this work relies in the use of a multimodal representation approach, specifically through the application of CLIP embeddings. By

encoding both textual and visual content into one shared space, allows the model to examine fashion data more completely than single modalities. Fashion thrives on intertwined visuals, captions, and styles, and the outcomes prove this method enables the identification of unified fashion types across formats, building on previous research that also affirm vision language models for capturing complex semantic relationships (Radford et al., 2021; Liu and Luo, 2020). Thus, the methodology offers a scalable solution for analysing large volumes of heterogeneous social media data without requiring manual feature engineering.

In comparison to traditional methods, the study illustrates the potential of unsupervised learning in uncovering patterns that are not predefined. Usual fashion studies lean on labelled data or preset labels, which restrict their ability to capture evolving or unexpected trends. However, the clustering approach used in this study lets structures surface organically from raw inputs. Clusters like promotional posts, resale scene, and niche styles reveal how these tools expose familiar and subtle patterns in social media. This aligns with wider developments in machine learning, where representation learning and clustering are increasingly used for exploratory analysis in complex, unlabelled datasets (Caron et al., 2018). Social media emerges here as more than a mirror of fashion currents but propels them too. Clusters heavy on promotion and influencers point to algorithms and commerce which controls the visibility of users and affects their choices. This reinforces the idea that trends identified through social media analysis must be interpreted within the context of platform dynamics, where engagement, sponsorship, and recommendation systems influence what becomes “Trendy” (Binns and Veale, 2023). Despite this, the alignment between several identified clusters and 2016 trends confirm that meaningful insights can still be extracted from such environments.

Overall, these insights prove that a data-driven, multimodal, unsupervised pipeline can effectively identify and track fashion trends within a complex and rapidly evolving digital environment. The ability to detect emerging patterns early, without a reliance on labelled data, represents a valuable contribution to both academic and practical research in the fashion industry. At the same time, the results highlight the limitations that follow, such as platform bias and data limitations when interpreting trends derived

from social media, which further suggests pathways for refinement and development in future work.

5.3 Limitations and Improvements

Although the proposed methodology demonstrates the ability to identify and analyse fashion trends from social media data, several limitations must be acknowledged. These constraints emerge from the inherent nature of the dataset and the methodological choices made in this study, as they have direct impact for interpretation and extending the results. The primary limitation involves the use of the Pinterest dataset, which introduces structural bias in the overall analysis. As noted in prior sections, Pinterest's content draws heavily from its recommendation algorithms, favouring visually similar and trending images. This results in a relatively homogeneous set of visual inputs, thereby diminishing the variety of fashion patterns accessible for clustering. Evidence of this bias appears in the clustering results, where certain visual themes are overrepresented, potentially limiting the model's ability to capture the full spectrum of fashion styles. Thus, the resulting clusters likely reflect platform-specific curation rather than a fully representative view of fashion trends. A further limitation arises from the lack of ground truth labels. Unsupervised learning permits the discovery of latent patterns without predefined labels, however, it also restricts the ability to quantitatively validate the semantic correctness of the clusters. Qualitative review through exemplar analysis indicated that the clusters hold interpretive value, yet no objective benchmark against clustering performance can be measured. This makes it difficult to assess the accuracy of the identified fashion archetypes beyond visual inspection and introduces an element of subjectivity into the evaluation process. The dataset's temporal boundaries further restrict the analysis, as the brief timeframe limits the depth of the trend analysis and reduces the reliability of the forecasting stage. Fashion trends are inherently seasonal and often evolve over extended periods, but the dataset offers merely a narrow glimpse into such processes. As a result, the time series used for forecasting were short and highly volatile, which restricted the ability of statistical models to learn stable patterns. This limitation is reflected in the forecasting results, where even simple baseline methods perform competitively, indicating a lack of consistent temporal structure.

In addition, the selection of forecasting models further constraints. The use of linear regression and a naïve baseline emphasise clarity and ease, nevertheless, these approaches are not well suited in addressing the intricate, nonlinear shifts common in social media trends. The linear model's failure to surpass the naïve baseline highlights the limitations of assuming a linear relationship in highly irregular data. While this choice is justified within the scope of the study, it restricts the predictive capability of the system and limits the extent to which meaningful forecasts can be generated. These limitations suggest multiple paths for refinement, expanding the dataset to include additional platforms like TikTok or full Instagram image data, would provide a more diverse and representative view of fashion content. Secondly, extending the datasets temporal range of the dataset would enable the analysis and season patterns, thereby improving the reliability of both trend detection and forecasting. From a methodological perspective, the application of more advanced time-series models could enhance predictive performance. Techniques such as ARIMA (AutoRegressive Integrated Moving Average) or recurrent neural networks (e.g. LSTM) would be better equipped to model temporal dependencies and non-linear behaviour and could provide more accurate forecasts given sufficient data. ARIMA was not used in this study mainly due to the short and highly volatile nature of the dataset, which violates key assumptions of stationarity and temporally stability required for reliable parameter estimation. Furthermore, incorporating techniques to address bias within the dataset, such as re-sampling strategies or adjustment weight, could help reduce the influence of dominant content and produce a more balanced clustering process.

Overall, while the methodology provides a structures and effective structure for fashion trend analysis, these limitations reveal the hidden hurdles of social media handling. Addressing them in future work would improve both the durability and utility, permitting deeper, and more precise analysis of fashion trends in digital environments.

5.4 Chapter Summary

This chapter examines the results in detail, the findings demonstrate that fashion trends can be effectively identified from multimodal social media data when combined with temporal analysis, and reveals the patterns to be found are consistent with real-world behaviour. The results also indicate that these trends are very dynamic and

volatile and therefore predictive modelling is not necessarily reliable. The method is useful for identification of trends and exploratory analysis, but forecasting remains a significant challenge within this context. Taken together, these observations reveal the methods strengths and its constraints, providing a balanced understanding of its practical applicability. The following chapter builds on these findings to present the overall conclusions of the study and outlines the direction for future work.

6 Conclusion and Future Work

6.1 Conclusion

This study's goal was to determine if largescale, multimodal datasets from social media could effectively identify and predict fashion trends. The problem was inherently challenging due to the unstructured, dynamic, and weakly labelled nature of social media content, where trends emerge organically rather than following clearly defined patterns. To address this, a data-driven pipeline was developed, which combined contrastive learning for multimodal representations through CLIP, unsupervised clustering, temporal aggregation, and simple forecasting models. The findings demonstrate that social media data does capture fashion trends accurately. The clustering stage revealed coherent categories such as Promotional Fashion Content, Menswear Tailoring, Denim Styles, and Luxury aesthetics. These clusters showed strong alignment with real-world fashion trend reported in 2015 – 2016, which confirmed that the multimodal embedding approach successfully captures meaningful semantic relationships across both textual and visual data. This highlights the capability of unsupervised learning methods to uncover latent structures within large, unlabelled datasets. Fashion archetypes on social media have been proved to be unstable over time as their environment is highly dynamic. Cluster shares fluctuate dramatically monthly, with some trends showing rapid growth followed by a sharp decline, whilst others maintained relatively stable presence. The use of momentum-based measures proved to be effective in identifying emerging trends, demonstrating that relative growth provides a stronger signal than absolute popularity in this field. However, the results also highlight clear limitations within prediction. The forecasting stage showed that a naïve baseline consistently outperformed linear regression, indicating that past trends do not reliably predict future behaviour within this context. This suggests that fashion trends on social media are influenced by unpredictable

external factors such as virality, influencer activity, and platform algorithms, making them inherently difficult to model using traditional time-series approaches.

To conclude, the study provides a flexible foundation for multimodal fashion analysis. It demonstrates that while social media data is highly effective for detecting and analysing trends, its value for forecasting remains limited. This distinction represents a key insight, emphasising that data-driven approaches are better suited for understanding current and emerging trends rather than predicting their future trajectory.

6.2 Future Work

While the proposed methodology establishes a strong foundation for analysing fashion trends, several opportunities exist for further improvement. These opportunities arise directly from the limitations identified in the previous chapter and reflect both data-related and methodological improvements. A primary area for improvement lies in expanding the dataset, as the current study relies on Instagram captions and a relatively small Pinterest image dataset, which introduces both imbalance and platform-specific bias. Future work could incorporate additional platforms such as TikTok, which plays a significant role in contemporary trend developments, as well as a full Instagram image dataset. Including a broader range of sources could improve the diversity and representativeness of the dataset, enabling a more comprehensive trend analysis. Furthermore, extending the timeframe to span multiple years would allow for the identification of seasonal patterns and long-term trends, which are not captured within the current short time frame. From a modelling perspective, more advanced analytical techniques could be explored, in terms of clustering, alternative methods such as density-based clustering (e.g. HDBSCAN) or hierarchical approaches may provide improvements in dividing fashion patterns, especially in noisy datasets. For trend forecasting, more sophisticated time-series models like ARIMA or LSTM networks would improve precision by detecting nonlinear relationships and temporal dependencies. As these models are well suited to capturing non-linear dependencies and temporal dynamics, which aren't handled effectively with linear regression.

Addressing bias within the dataset also represents an important direction for future work. Techniques such as re-sampling, weighting, or cross-platform validation could be applied to reduce the influence of overrepresented content, particularly from recommendation-driven platforms like Pinterest. These measures would strengthen cluster validity and broaden applicability. In addition, the evaluation methods could be improved as well, as unsupervised techniques alone invite subjective interpretations. Whereas introducing labelled datasets or expert assessments provides objective benchmarks for assessing cluster coherence and trend identification. This would strengthen the validation of unsupervised learning outcomes and reduce reliance on qualitative interpretation alone.

In summary, future work should focus on enhancing data diversity, extending temporal coverage, applying more advanced modelling techniques, and improving evaluation and system deployment. These developments would significantly strengthen the capability of the framework, moving it closer to a stronger and practical tool for real-world fashion trend analysis.

7 Reflection

This project has been a valuable learning experience, not only in terms of technical development but also in understanding how to approach complex, real-world problems in a structured and critical manner. The main aim was to explore whether fashion trends could be identified and predicted using social media data. While the technical implementation was important, the process of refining this problem, selecting appropriate methods, and evaluating the results proved to be equally significant in shaping my overall learning. One of the key skills developed during this project was the ability to design and implement a complete data-driven pipeline. This included working with multimodal data, applying machine learning techniques such as clustering, and interpreting time-series behaviour. More importantly, I developed a stronger understanding of how different components of a system interact, particularly how preprocessing, representation learning, and modelling decisions directly influence the quality of results. The use of CLIP and unsupervised learning also deepened my appreciation for modern machine learning approaches that rely less on manual feature engineering and more on representation learning.

A major challenge encountered throughout the project was dealing with the limitations of real-world data. The datasets were noisy, imbalanced, and in some cases biased due to platform-specific characteristics. In particular, the Pinterest dataset introduced a level of homogeneity that constrained the diversity of visual patterns, while the lack of labelled data made it difficult to objectively evaluate clustering performance. Additionally, the short time horizon of the dataset limited the effectiveness of forecasting models, highlighting the difficulty of working with incomplete or imperfect data. While some of these challenges were addressed through preprocessing and careful interpretation, others remained unresolved and influenced the overall outcomes of the project.

Reflecting on the project, one area that could be improved is the initial planning of the data collection process. If approached again, I would prioritise obtaining a larger and more diverse dataset, particularly by incorporating additional platforms and extending the time range. This would provide a stronger foundation for both trend analysis and forecasting. Furthermore, I would explore more advanced modelling techniques earlier in the process, allowing for a more comprehensive comparison of methods rather than focusing primarily on simpler baseline models. The project highlighted the importance of continuously evaluating assumptions and adapting the approach based on findings. For example, while the initial objective included forecasting fashion trends, the results demonstrated that prediction was significantly more challenging than detection. This required a shift in focus towards understanding why forecasting was difficult, rather than attempting to force improved predictive performance.

Overall, this project has strengthened my ability to approach open-ended problems, critically evaluate results, and recognise the limitations of data-driven methods. It has also provided insight into the practical challenges of applying machine learning in real-world contexts, where data quality, bias, and unpredictability play a significant role. These experiences will be valuable for future work, particularly in developing more robust and adaptable analytical systems.

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Appendices

Appendix A

Dashboard Screenshots

The following screenshots demonstrate the implementation of the interactive Streamlit dashboard developed for fashion trend exploration, forecasting, and cluster interpretation.

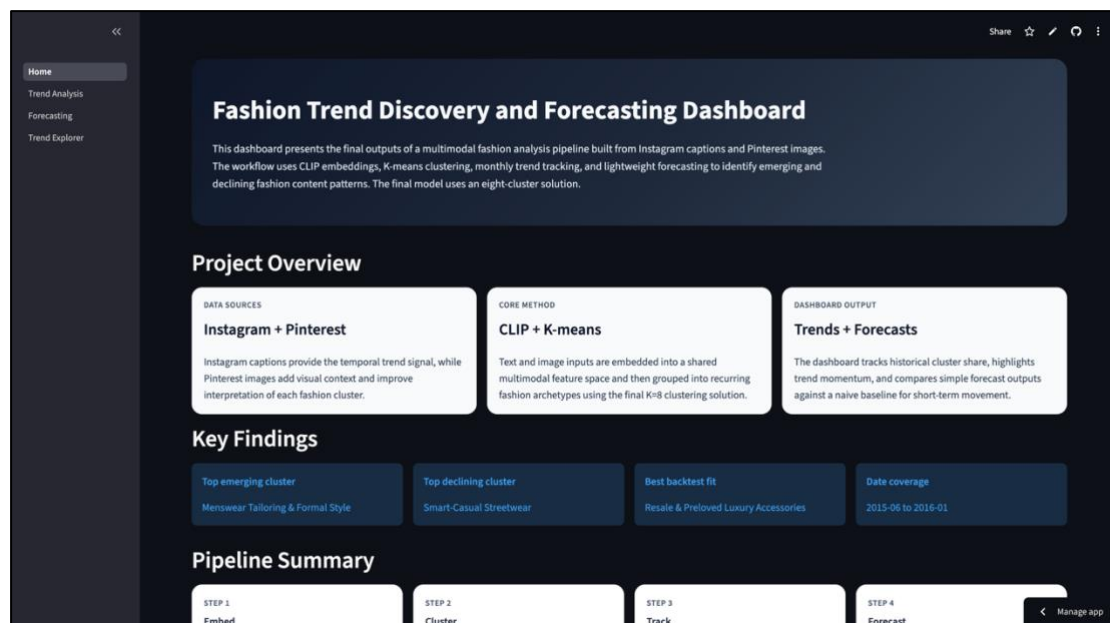


Figure A.1: Streamlit dashboard home page presenting project overview and navigation structure.

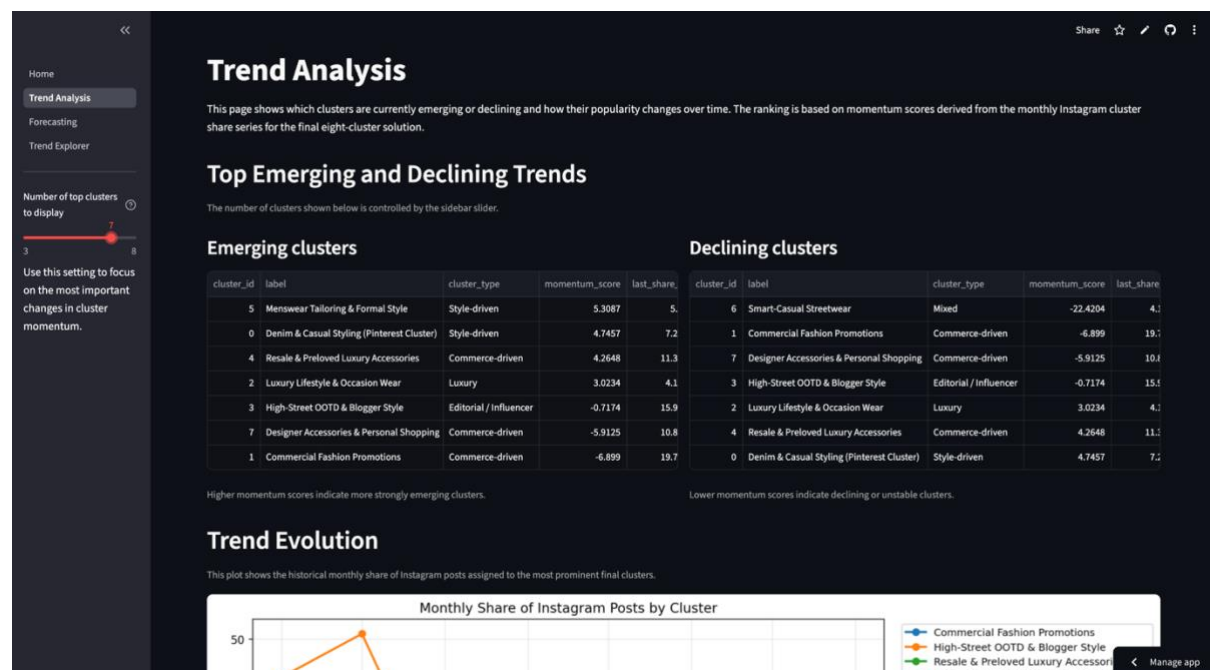


Figure A.2: Trend Analysis page showing monthly evolution of fashion cluster popularity over time and emerging and declining clusters.

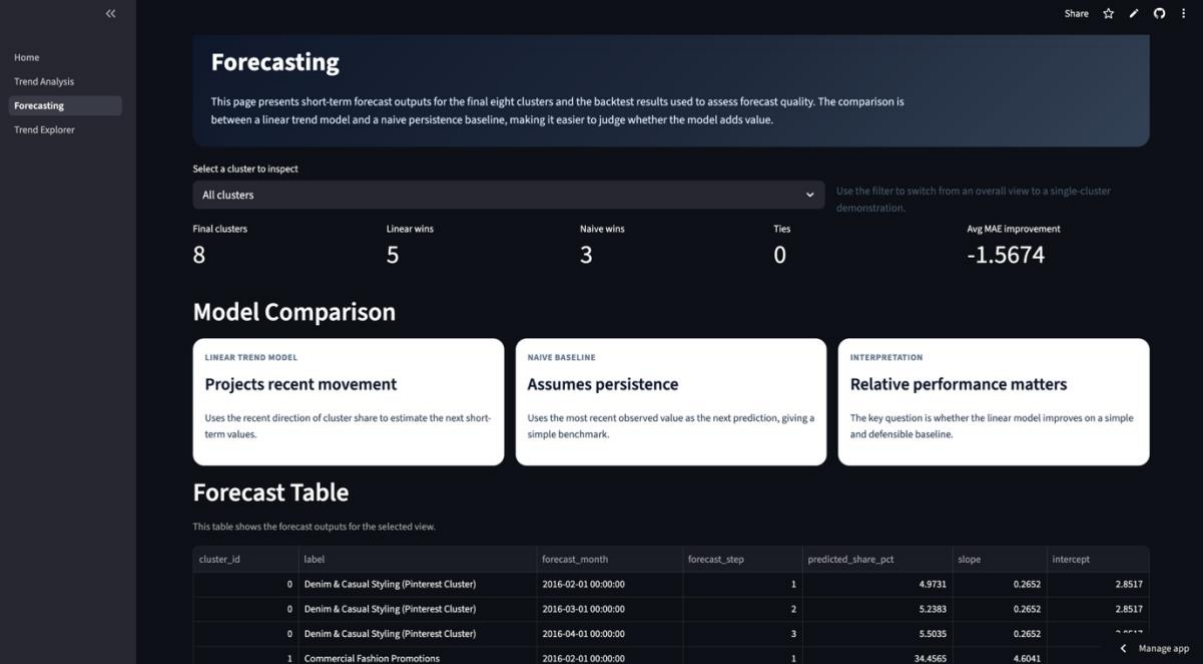


Figure A.3: Forecasting dashboard displaying linear regression forecasts against historical trend data.

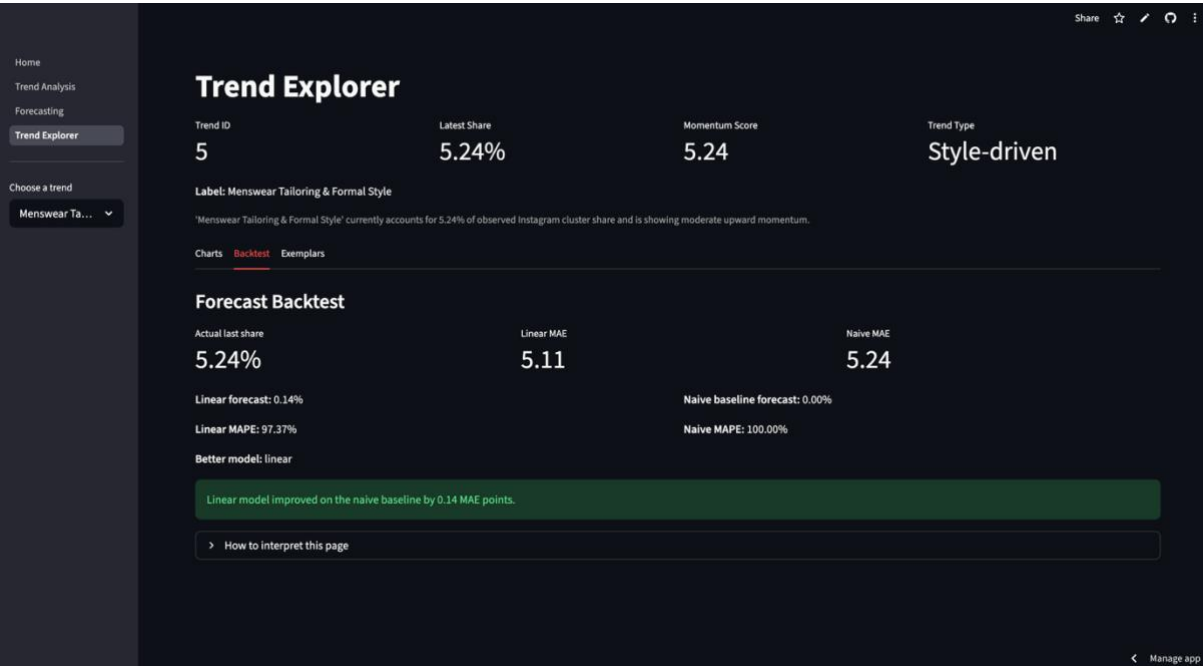


Figure A.4: Forecast backtesting interface comparing model predictions against naive baseline forecasts on the Trend Explorer page.

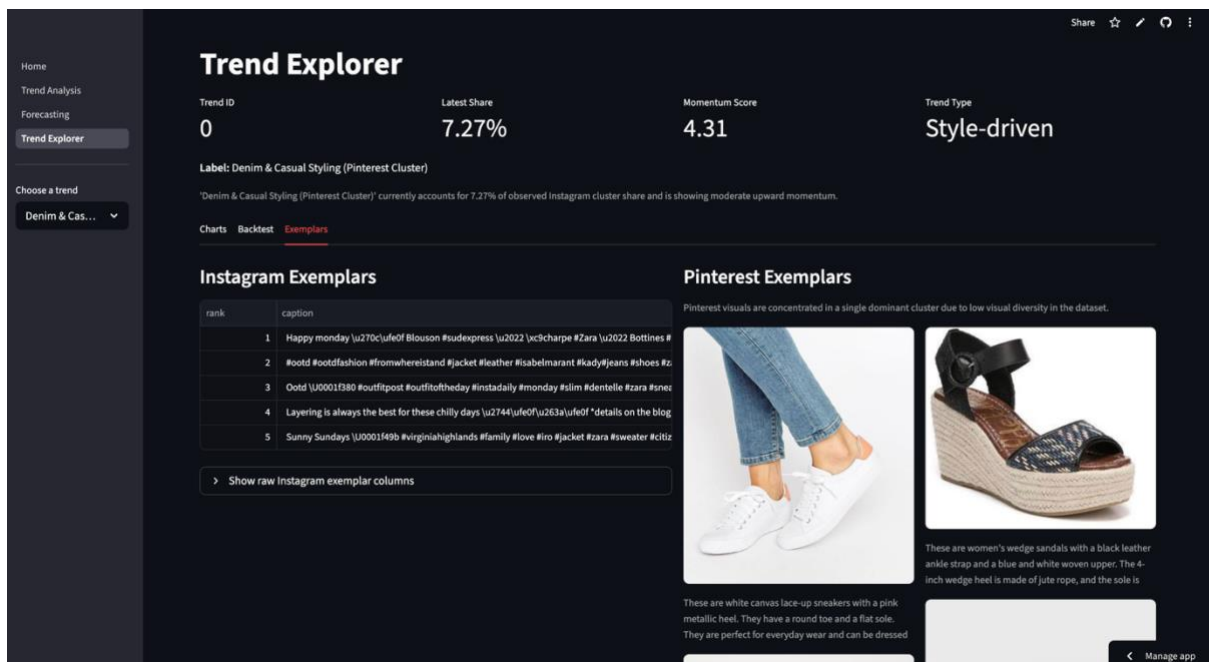


Figure A.5: Trend Explorer page visualising cluster exemplar images and representative Instagram captions.



Figure B.6: Interactive cluster filtering interface used for detailed fashion archetype exploration.

Appendix B

Forecasting Evaluation Tables

The following tables present supplementary forecasting evaluation results referenced throughout Chapters 4 and 5. These outputs provide additional detail regarding forecasting accuracy, model comparison, and backtesting performance across the discovered fashion clusters. The appendix supports the discussion of predictive reliability and highlights the comparative performance between the linear regression model and the naïve baseline approach.

cluster_id	label	forecast_month	forecast_step	predicted_share_pct	slope	intercept
0	Denim & Casual Styling (Pinterest Cluster)	2016-02-01 00:00:00	1	4.9731	0.2652	2.8517
0	Denim & Casual Styling (Pinterest Cluster)	2016-03-01 00:00:00	2	5.2383	0.2652	2.8517
0	Denim & Casual Styling (Pinterest Cluster)	2016-04-01 00:00:00	3	5.5035	0.2652	2.8517
1	Commercial Fashion Promotions	2016-02-01 00:00:00	1	34.4565	4.6041	-2.3766
1	Commercial Fashion Promotions	2016-03-01 00:00:00	2	39.0606	4.6041	-2.3766
1	Commercial Fashion Promotions	2016-04-01 00:00:00	3	43.6647	4.6041	-2.3766
2	Luxury Lifestyle & Occasion Wear	2016-02-01 00:00:00	1	1.6531	-0.1703	3.0159
2	Luxury Lifestyle & Occasion Wear	2016-03-01 00:00:00	2	1.4827	-0.1703	3.0159
2	Luxury Lifestyle & Occasion Wear	2016-04-01 00:00:00	3	1.3124	-0.1703	3.0159
3	High-Street OOTD & Blogger Style	2016-02-01 00:00:00	1	0	-5.388	40.0991
3	High-Street OOTD & Blogger Style	2016-03-01 00:00:00	2	0	-5.388	40.0991
3	High-Street OOTD & Blogger Style	2016-04-01 00:00:00	3	0	-5.388	40.0991
4	Resale & Preloved Luxury Accessories	2016-02-01 00:00:00	1	11.9311	1.0002	3.9299
4	Resale & Preloved Luxury Accessories	2016-03-01 00:00:00	2	12.9313	1.0002	3.9299
4	Resale & Preloved Luxury Accessories	2016-04-01 00:00:00	3	13.9314	1.0002	3.9299
5	Menswear Tailoring & Formal Style	2016-02-01 00:00:00	1	2.6924	0.4269	-0.7233
5	Menswear Tailoring & Formal Style	2016-03-01 00:00:00	2	3.1194	0.4269	-0.7233
5	Menswear Tailoring & Formal Style	2016-04-01 00:00:00	3	3.5463	0.4269	-0.7233
6	Smart-Casual Streetwear	2016-02-01 00:00:00	1	21.601	0.9026	14.3806
6	Smart-Casual Streetwear	2016-03-01 00:00:00	2	22.5036	0.9026	14.3806
6	Smart-Casual Streetwear	2016-04-01 00:00:00	3	23.4061	0.9026	14.3806
7	Designer Accessories & Personal Shopping	2016-02-01 00:00:00	1	16.6577	1.6274	3.6386
7	Designer Accessories & Personal Shopping	2016-03-01 00:00:00	2	18.2851	1.6274	3.6386
7	Designer Accessories & Personal Shopping	2016-04-01 00:00:00	3	19.9125	1.6274	3.6386

Figure B.1: Forecast Table of all clusters with a short time estimate of their projection.

cluster_id	label	mae_linear	mae_naive	mae_improvement_vs_naive
0	Denim & Casual Styling (Pinterest Cluster)	4.4002	4.7229	0.3227
1	Commercial Fashion Promotions	17.2665	21.8466	4.5801
4	Resale & Preloved Luxury Accessories	0.6496	5.4088	4.7592
5	Menswear Tailoring & Formal Style	5.1059	5.244	0.138
7	Designer Accessories & Personal Shopping	7.139	10.3462	3.2072

Figure B.2: Clusters where the linear model beats the naïve baseline

cluster_id	label	actual_last_share_pct	predicted_last_share_pct_linear	predicted_last_share_pct_naive	mae_linear	mae_naive	mape_pct_linear	mape_pct_naive	better_model	mae_improvement_vs_naive
0	Denim & Casual Styling (Pinterest Cluster)	7.2747	2.8745	2.5518	4.4002	4.7229	60.4863	64.9219	linear	0.3227
1	Commercial Fashion Promotions	19.7802	37.0467	41.6268	17.2665	21.8466	87.2917	110.4466	linear	4.5801
2	Luxury Lifestyle & Occasion Wear	4.1846	0.1368	0.7974	4.0478	3.3872	96.7298	80.9433	naive	-0.6606
3	High-Street OOTD & Blogger Style	15.9209	0	5.5821	15.9209	10.3387	100	64.9383	naive	-5.5821
4	Resale & Preloved Luxury Accessories	11.3099	10.6603	5.9011	0.6496	5.4088	5.7432	47.8234	linear	4.7592
5	Menswear Tailoring & Formal Style	5.244	0.138	0	5.1059	5.244	97.3679	100	linear	0.138
6	Smart-Casual Streetwear	4.1187	32.5412	13.2376	28.4225	9.119	690.0868	221.4048	naive	-19.3035
7	Designer Accessories & Personal Shopping	10.8659	18.0049	21.2121	7.139	10.3462	65.7007	95.2167	linear	3.2072

Figure B.3: Backtest results of all clusters